

Reviewed Preprint

v1 • August 13, 2025

Not revised

Reviewed Preprint

v2 • January 26, 2026

Revised by authors

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Competing interests:

No competing interests declared

Funding:

See [page 39](#)

Reviewing editor:

John W Schoggins, The University of Texas Southwestern Medical Center, United States

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A rapid transfer of virions coated with heparan sulfate from the ECM to CD151 defines an early step in the human papillomavirus infection cascade

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eLife Assessment

This study provides **valuable** insight into the role of actin protrusions in mediating early pre-endocytic steps of human papillomavirus entry at the cell surface. Using state-of-the-art microscopy in an immortalized keratinocyte model, the authors present mostly **solid** evidence that filopodia actively promote the transfer of heparin sulfate-coated virions from the extracellular matrix to the viral entry factor CD151. Remaining gaps in the mechanistic model could be further supported by including a more expansive analysis of the fixed microscopy samples and live cell imaging to distinguish virion transfer from direct binding.

<https://doi.org/10.7554/eLife.107139.2.sa3>

Abstract

Human Papillomaviruses (HPVs) are the underlying cause of several types of cancer, albeit they are mostly known for their association with cervical carcinoma. The virions must reach their target cells through a break in the epithelial barrier. After binding to heparan sulfate (HS) of the extracellular matrix (ECM), they translocate to the cell surface and co-internalize with the entry factor CD151.

The *in vivo* occurring translocation from the ECM to entry factors at the cell membrane may be bypassed in nonpolarized cells with low ECM secretion. To specifically investigate these early events of the infection cascade, we use polarized keratinocytes. They produce a robust ECM and have a restricted expression of entry receptors primarily to the substrate-adhered basal membrane. Virion access to the basal membrane by diffusion is strongly limited, such that translocation from the ECM is the main pathway used for infection. We block the translocation from ECM attachment sites to the cell body, release the blocking, and monitor the association of virions with CD151 or HS. We observe quick virion translocation from the ECM to the cell body within 15 min. During translocation, virions associate with the tetraspanin CD151 present at the cell border or at filopodia. Translocating virions are decorated with HS, which they lose in the next few hours, presumably prior to endocytosis.

Our observations reveal a rapid step in the HPV infection cascade: the transfer of HS-coated virions from the ECM to CD151. This step is too fast to account for the asynchronous uptake of HPVs which is likely driven by glycan- and capsid processing.

Introduction

Already in the 80s, Harald zur Hausen proposed a role of human papillomaviruses in cancer (zur Hausen, 2009). Since then, five more classes of oncogenic viruses have been identified (Galati et al., 2024a). To date, it is assumed that more than 10% of the worldwide human cancer burden is associated with infectious agents (Galati et al., 2024a), from which about a half is caused by *Papillomaviridae* (Martel et al., 2017). Thus, the understanding of viral entry strategies has implications going beyond the classical treatment of acute viral infections.

Human papillomaviruses are small, non-enveloped viruses with a diameter of ≈ 55 nm. The icosahedral capsid is mainly composed of pentameric L1 capsomers. Together with less L2 capsid proteins, capsomers surround a histone core bearing a circular double-stranded DNA (Baker et al., 1991; Ozbun and Campos, 2021). From the more than 200 phylogenetically classified HPV genotypes, the most oncogenic ones are HPV16 and HPV18 (Galati et al., 2024b), which are responsible for about 70 - 80% of the cervical cancer cases (Christiansen et al., 2015). In addition, they cause other severe cancers such as anogenital, head and neck tumors (Doorbar et al., 2012).

For papillomavirus infection, a disruption of the epithelial barrier is a prerequisite, through which virions reach mitotically active basal cells of the epithelia (Ozbun, 2019). Here, virions bind to the linear polysaccharide heparan sulfate (HS) that is present in the extracellular matrix (ECM) but as well on the plasma membrane surface. HS is attached to proteins forming so called heparan sulfate proteoglycans (HSPGs). Positively charged and polar residues of the L1 capsid protein form multiple heparan sulfate binding sites that interact with negative charges of HS, resulting in a strong bond and virus uptake into the infectious pathway (Dasgupta et al., 2011; Girolglou et al., 2001; Joyce et al., 1999; Knappe et al., 2007; Surviladze et al., 2015). While in cell culture virions bind to HS on both the cell surface and the ECM, it has been suggested that *in vivo* they bind predominantly to HS of the extracellular basement membrane (Day and Schelhaas, 2014; Kines et al., 2009; Ozbun and Campos, 2021; Schiller et al., 2010). In any case, the tight bond between the linear polysaccharide and virions must be disrupted before they can bind to a yet unknown secondary receptor on the cell surface, followed by internalization (Ozbun and Campos, 2021).

The strong electrostatic bonding precludes HS release via dissociation. Two release mechanisms are discussed, that mutually are not exclusive. In the so-called priming model, binding to HS structurally results in capsid enlargement and softening (Feng et al., 2024), followed by the exposure and cleavage of L1 by kallikrein-8 (KLK8) (Cerqueira et al., 2015), exposure of L2 by cyclophilin (Bienkowska-Haba et al., 2009), and L2 cleavage by furin (Richards et al., 2006). After these structural modifications of the capsid surface the 'primed' virion is able to bind to the secondary receptor. In an alternative model, the HS-virion bond remains. However, cleavage of HS/HSPGs by heparanases and proteinases produces HS fragments. Therefore, albeit still bound to the virion surface, the fragmented HS no longer anchors the virion to the ECM (Surviladze et al., 2012; Surviladze et al., 2015).

Once the virions are released, they could reach the cell surface simply by passive diffusion. In a co-culture trans-well assay, pseudovirions (PsVs) from donor cells infect spatially separated receiver cells (Surviladze et al., 2012), demonstrating that free diffusion of virions may be sufficient for infection. However, an active transport mechanism has been reported as well. PsVs migrate along actin-rich protrusions from the ECM towards the cell body (Schelhaas et al., 2008; Smith et al., 2008). While it is generally agreed on that HS is a primary virion attachment site, the molecular identity of the secondary receptor on the main cell body surface is unknown. The secondary complex is most likely a multimeric complex rather than a single molecular component. Possible candidates are proteins crucial for cell entry, as the tetraspanin CD151 (Mikulić et al., 2019; Scheffer et al., 2013; Spoden et al., 2008), integrin- $\alpha 6$ (Itga6) (Evander et al., 1997; Yoon et al., 2001), growth factor receptors (Mikulić et al., 2019; Surviladze et al., 2012), and the annexin A2 heterotetramer (Dziduszko and Ozbun, 2013; Woodham et al., 2012). From these molecules, the tetraspanin CD151 could play a coordinating role, as tetraspanins organize viral entry platforms in many types of viral infections, including infections with coronavirus,

cytomegalovirus, hepatitis C virus, human immunodeficiency virus, human papilloma virus, and influenza virus (Bruening et al., 2018 [↗](#); Earnest et al., 2015 [↗](#); Florin and Lang, 2018 [↗](#); Hantak et al., 2019 [↗](#); Hochdorfer et al., 2016 [↗](#); Scheffer et al., 2013 [↗](#); Zona et al., 2013 [↗](#)). Tetraspanin entry platforms could form in a slow and stochastic fashion, which would provide an explanation for the asynchronous virion uptake with half-times of above 10 h (Becker et al., 2018 [↗](#)). However, it is unclear whether virions associate with CD151 already in the moment of virion transfer from the ECM to the cell surface or afterwards in preparation of the endocytic uptake.

In this study, we explore the relationship between actin-dependent transport and CD151 association in HPV infection. We employ the cell permeable mycotoxin and actin polymerization inhibitor cytochalasin D (CytD). We find that CytD causes the trapping of HPV16 PsVs in the ECM of a polarized human keratinocyte cell line (HaCaT cells), where they remain accumulated adjacent to the cell periphery. This blocking of the virus transfer is accompanied by co-accumulation of HS in the ECM area. Upon CytD removal, HS-decorated PsVs move from the ECM to the cell body and CD151. The association of PsVs with CD151 persists within the next few hours, whereas the HS coat is stripped off. These findings distinguish a step in the infection cascade with CD151 playing an early role directly after the release of virions from the ECM.

Results

Cytochalasin D arrests the translocation of HPV16 PsVs from the extracellular matrix to the cell body surface

The molecular surface of PsVs is immunologically indistinguishable from HPV virus particles (Ozbun and Campos, 2021 [↗](#)), which makes them a widely used tool for studying host cell entry. In this study, we employ HPV16 pseudovirions with an encapsidated luciferase reporter plasmid under the control of the HPV16 promotor (Wüstenhagen et al., 2018 [↗](#)). Hence, instead of viral DNA a luciferase encoding plasmid enters the nucleus, enabling the analysis of the infection rate via the luciferase activity. Moreover, the plasmids are composed of nucleotides to which fluorophores can be coupled by click-chemistry. This allows for their detection by fluorescence microscopy throughout the entire infection pathway.

Many PsVs that bind to the ECM may locate distal from the cell surface and are thus unable to establish direct contact with entry receptors. However, they are capable of migrating by an actin-dependent transport along cell protrusions towards the cell body (Schelhaas et al., 2008 [↗](#); Smith et al., 2008 [↗](#)). We aimed for blocking this transport in HaCaT cells, a cell line that is widely used as a cell culture model for HPV infection. HaCaT cells closely resemble primary keratinocytes in key aspects: they are not virally transformed and produce large amounts of ECM that facilitates infection (Bienkowska-Haba et al., 2018 [↗](#); Gilson et al., 2020 [↗](#)). In addition, HaCaT cells exhibit cellular polarity that enforces binding of virus particles to the ECM, as the virions cannot bind to receptors/entry components, such as CD151, Itga6 and HSPGs that co-distribute on the basolateral membrane of polarized keratinocytes (Cowin et al., 2006 [↗](#); Mertens et al., 1996 [↗](#); Sterk et al., 2000 [↗](#)), making them inaccessible by diffusion.

We incubate cells for 5 h with PsVs in the absence or presence of 10 µg/ml CytD. Previously, it has been shown that this CytD concentration stops cell migration after a few minutes (Peng et al., 2011 [↗](#)). Frequently, we observe patches of confluent cells which are common to HaCaT cells. Cells at the center of these patches are dismissed during imaging, because there are no anterogradely migrating PsVs at these cells. A second reason for our dismissal of these cells is that hardly any PsVs are bound to them, possibly because their basal membranes are inaccessible by diffusion. Instead, we focus on isolated HaCaT cells or cells at the periphery of cell patches. At these cells, we find more PsVs per cell than one would expect from the employed 50 viral genome equivalents (vge) per cell, indicating that PsVs are unequally distributed between the cells. Moreover, these PsVs usually are not homogenously distributed around the cell but concentrate at one region. We investigate the translocation of PsVs from these regions, defining ROIs for analysis that cover PsVs at the periphery and the cell body (see Supplementary Figures 6A [↗](#) and 8A [↗](#)). After the 5 h incubation, cells were fixed and stained by a membrane marker, the dye TMA-DPH, to mark the

main cell body (Figure 1A [↗](#), gray lookup table (LUT)). Additionally, they were stained for PsVs (by click-chemistry; magenta LUT), HS (antibody staining; cyan LUT), and the membrane protein Itga6 (antibody staining; green LUT). In the control, PsVs locate mostly in the cell body area (Figure 1A [↗](#), upper row; see white outline that is based on TMA-DPH staining). In contrast, when CytD was present, the cell body area is largely devoid of PsVs (Figure 1A [↗](#), lower row; please note that the large central bright area is mainly caused by autofluorescence). Instead, PsVs often accumulate adjacent to the cell body. This adjacent region that is up to several μm wide and rich in HS, likely contains the ECM. We conclude that, upon inhibition of actin-dynamics by CytD, PsVs remain accumulated in the ECM area.

Comparing in Figure 1A [↗](#) the PsV brightness in the upper (control, magenta LUT) and lower (CytD, magenta LUT) panels suggests that the amount of PsVs does not differ greatly. This is surprising, as CytD is not reported to inhibit enzymes involved in capsid modification or HS/HSPGs processing, after which the PsVs supposedly leave the ECM. If this were the case, one would expect no difference in the number of PsVs at the periphery between cells that were treated with or without CytD for 5 h. Therefore, we conclude that PsVs remain ECM associated while ready for binding to a secondary receptor. However, a quantification of the PsVs with and without CytD is not possible in this experiment, due to interference from autofluorescence (Figure 1A [↗](#), large magenta area in the center). To study PsVs in the absence of autofluorescence, we employed unroofed cells (Heuser, 2000 [↗](#)), also referred to as membrane sheets. After pre-treatment of cells as above, membrane sheets are generated by brief ultrasound pulses that remove the upper cellular parts (and thus any intracellular autofluorescence), leaving behind the extracellular matrix and the basal membrane along with the bound PsVs. During imaging, membrane sheets are identified via staining of F-actin with fluorescently labelled phalloidin (Figure 1B [↗](#), green LUT) that marks the main cell body cortex together with lamellipodia and filopodia. In this experiment we observe that PsVs, not always but frequently, accumulate adjacent to the isolated basal membrane (defined by F-actin staining, Figure 1B [↗](#), green LUT) after CytD treatment, which differs from the control (Figure 1B [↗](#), magenta LUT).

The PsV maxima density (Figure 1B [↗](#), magenta LUT, detected as local maxima of antibody stained L1) and the PsV intensity are quantified in an area covering the cell body and the periphery. Compared to the control, CytD reduces the PsV maxima density by 26% (the PsV maxima density in the control and CytD treated cells is 0.7 and 0.52 PsVs/ μm^2 , respectively), whereas the maxima intensity increases by 41% (Figure 1C [↗](#)). A histogram of the PsV maxima intensities illustrates that CytD broadens the intensity distribution towards brighter PsVs (Figure 1D [↗](#)). The increase of the PsV intensity after CytD treatment can also be appreciated comparing the upper and middle/lower L1 image of Figure 1B [↗](#) (shown at the same settings of brightness and contrast). As CytD is unlikely to act directly on PsVs (e.g., by fusing them to larger and brighter particles), we assume that the underlying cause of brighter maxima is the resolution limit of the microscope. The densely accumulated PsVs are no longer resolved and merge to brighter spots. These poorly resolved maxima are on average 41% brighter but of them there are 26% fewer. The decrease in maxima density is in the order of magnitude of the increase in intensity, which yields a roughly similar total signal in both conditions.

We addressed whether PsVs would still be accumulated when adding CytD 1 h after the PsVs rather than simultaneously (Figure 1B [↗](#), lower panel, CytD after 1 h). As in the CytD condition, we frequently see accumulated PsVs adjacent to the isolated basal membrane (Figure 1B [↗](#), arrows point towards PsV accumulations). The PsV maxima intensity distribution is broader when compared to the control but narrower than with CytD present the entire time (Figure 1D [↗](#)). We conclude that complete PsV binding and translocation takes longer than 1 h, as otherwise control and CytD after 1 h should be more similar.

At this point, one could conclude from the data that CytD arrests PsVs on their journey towards the cell body and that PsVs are not collected from the cells, but remain trapped in the ECM. How efficient PsV trapping is cannot be determined exactly for several reasons. First, our assay is not precise for technical reasons, as the above-mentioned limited resolution results in an underestimation of the number of PsVs the stronger they are accumulated. Second, we do not

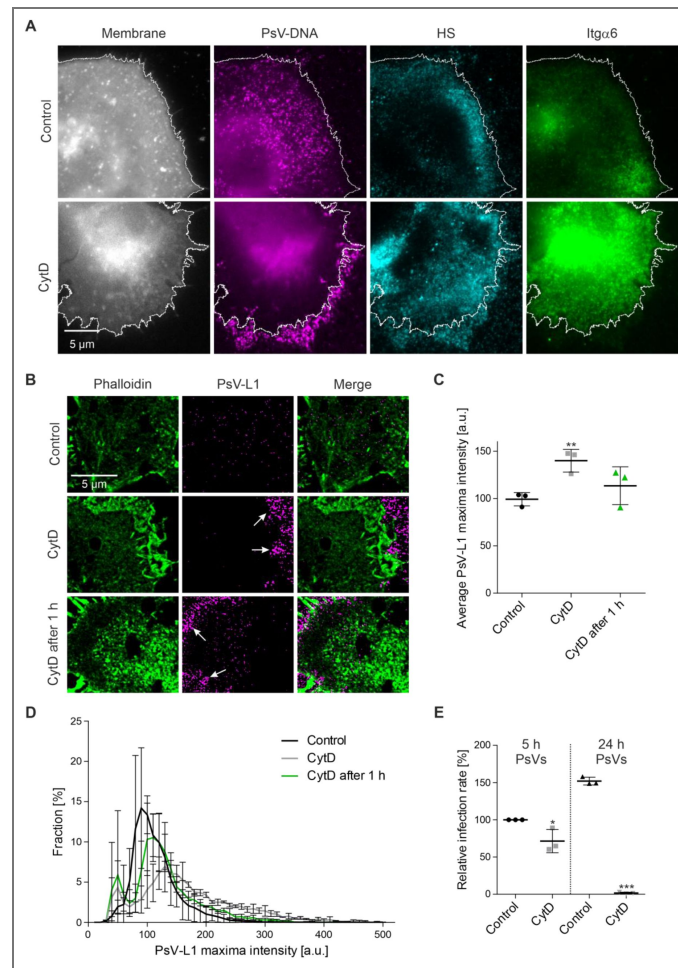


Figure 1. CytD arrests PsV translocation from the ECM to the cell body.

(A) In the absence (Control) or presence of 10 μ g/ml CytD (CytD), HaCaT cells were incubated with PsVs at 37 $^{\circ}$ C for 5 h. Then, cells were fixed, washed and stained with the cell membrane dye TMA-DPH (gray lookup table (LUT)). PsVs (magenta LUT) were visualized through coupling a dye (6-FAM Azide) to the encapsidated plasmid by click-chemistry. Indirect immunolabeling was employed for staining of HS (AlexaFluor 594; cyan LUT) and Itga6 (STARRED; green LUT). Imaging was realized with epi-fluorescence microscopy. White lines delineate the main cell body; lines were created with reference to the TMA-DPH membrane staining. (B) Same pre-treatment of cells as in (A), with an additional condition where CytD was added 1 h after the PsVs (CytD after 1 h). Prior to fixation, membrane sheets were generated and F-actin was stained with phalloidin coupled to iFluor488 (green LUT). The capsid protein L1 of the PsVs and CD151 were stained by immunofluorescence using primary antibodies in combination with AlexaFluor 594-labelled (L1, magenta LUT) and STAR RED-labelled (CD151, not shown in this figure for clarity reasons) secondary antibodies. Images of phalloidin and L1 staining were acquired in the confocal and STED mode of a STED microscope, respectively. Arrows in the PsV-L1 images point towards accumulated PsVs that after CytD are more frequently observed than in the control (see text). (C) Analysis of images as shown in (B) using ROIs covering the whole image. PsV maxima were detected and their intensities were quantified in a circular 125 nm diameter region of interest (ROI), followed by background correction. Values are given as means $\pm$ SD ($n = 3$; one biological replicate includes per condition the average of 14 - 15 analyzed membranes (intensity values of one membrane sheet were averaged) with altogether at least 1000 maxima intensity values). (D) PsV maxima intensity distribution of the data in (C). The fraction of PsVs, expressed in percent, is plotted as histogram (10 a.u. bins) against the maxima intensity. Values are given as means $\pm$ SD ($n = 3$). (E) HaCaT cells were treated either for 5 h or 24 h with PsVs, together without (Control) or with 10 μ g/ml CytD (CytD). In case of the 5 h incubation, after removal of the PsVs/CytD cells were incubated for another 19 h in medium (in total 24 h). After a total of 24 h incubation, the luciferase activity of lysed cells was measured, yielding the infection rate that was normalized to LDH, resulting in the normalized infection rate. The normalized infection rate was further related to the mean normalized infection rate of the 5 h control, set to 100%, yielding the relative infection rate. Values are given as means $\pm$ SD ($n = 3$ biological replicates; the value of one biological replicate is the average of three technical replicates). (C) and (E). Statistical differences between Control and CytD was analyzed by using the two-tailed, unpaired student's t -test ($n = 3$, for details see methods). a.u., arbitrary units.

know how many PsVs in the control become endocytosed during the 5 h of incubation. This reduces the PsVs in the control, which is our value for comparison. Moreover, it should be noted that both the degree of PsV accumulation as well as the number of PsVs are highly variable. Even in the absence of CytD, we occasionally find cells with accumulated PsVs at the periphery, whereas such cells are more frequently observed after CytD. This causes a large variability within the data. Still, we consider larger PsV accumulations after CytD as a typical effect (please note that for illustration of this effect we show in the Figures such examples). Despite these limitations, it appears that within the 5 h incubation period with CytD a large fraction of PsVs remains associated with the ECM.

During the CytD incubation, PsVs bind to HSPGs of the basolateral membrane for 5 h. Still, in the cell body area hardly any PsVs are present ($0.14 \text{ PsV}/\mu\text{m}^2$, Supplementary Figure 1B [↗](#)). In the control, the PsV density is several-fold larger (Supplementary Figure 1B [↗](#)). This is expected, as the PsVs bind to the ECM and translocate to the cell body. We wondered whether there are more binding sites at the basal membrane that, however, remain inaccessible to PsVs by diffusion, because of the insufficient space between glass-coverslip and basolateral membrane. For clarification, we incubated EDTA detached HaCaT cells in suspension with PsVs for 1 h at 4°C , followed by re-attachment for 1 h. Under these conditions, we find a PsV density 12.4-fold larger than after 5 h of CytD incubation of adhered cells (Supplementary Figure 1B [↗](#) and D [↗](#)). However, it should be noted that these values cannot be directly compared. Aside from the different treatments, another difference lies in the size of the basal membrane, as re-attachment of cells is not complete after only 1 h (compare size of adhered membranes in Supplementary Figure 1A [↗](#) and C [↗](#)). Therefore, the imaged membranes are likely strongly ruffled, which results in the underestimation of the size of the adhered membrane. As a result, we overestimate the PsVs per μm^2 (please note that we cannot re-attach cells for longer times as we would then lose PsVs due to endocytosis). On the other hand, we would underestimate the PsV density at the basal membrane if after re-attachment we image in part also some apical membrane. In any case, the experiment suggests that PsVs bind more efficiently if membrane surface receptors are accessible by diffusion. This is in support of the above notion that the basal membrane may provide more entry receptors than one would expect from the low density of PsVs bound after 5 h CytD (Supplementary Figure 1B [↗](#)). This suggests that under our assay conditions, PsVs cannot easily bypass the translocation from the ECM to the cell body by diffusing directly to the basal membrane. Hence, the large majority of PsVs that enter the cell were previously bound to the ECM. Therefore, HaCaT cells serve as an ideal model for studying the transfer of ECM bound HPV particles to the cell surface, which is similar to *in vivo* infection of basal keratinocytes after binding to the basement membrane (Bienkowska-Haba et al., 2018 [↗](#); Day and Schelhaas, 2014 [↗](#); Kines et al., 2009 [↗](#); Schiller et al., 2010 [↗](#)).

The blocking of PsV translocation by cytochalasin D is reversible

CytD not only arrests the transport of PsVs to the cell but is known to block other actin-dependent processes, which strongly affects the physiology of the cell. Therefore, we investigated whether PsVs would proceed normally on their infection pathway once CytD is washed off. In order to allow for recovery from CytD, cells were washed after the 5 h simultaneous PsV/CytD treatment, then incubated for another 19 h and the infection rate was subsequently determined by measuring the luciferase activity in the cell lysate. CytD reduced luciferase activity by 29%, as compared to the control (Figure 1E [↗](#); left). A certain degree of reduced infectivity is to be expected, as PsVs have $\approx 20\%$ less time to complete infection as compared to cells that were not treated with CytD. When PsVs were not washed off after 5 h but left for the entire 24 h incubation, the infection rate increased by 52%, whereas continuous treatment with CytD results in no infection (Figure 1E [↗](#); right). The latter is in line with previous studies (Schelhaas et al., 2008 [↗](#); Selinka et al., 2002 [↗](#); Selinka et al., 2007 [↗](#); Spoden et al., 2013 [↗](#)) showing that CytD is a strong inhibitor of HPV infection. In any case, PsVs proceed on the infection pathway upon removal of CytD. The reduced infection rate of 29% can be explained by the 5 h blocking at the beginning of

the incubation period that should delay the time course of infection. Therefore, we propose that CytD is suitable for transiently arresting PsVs in a state between primary attachment to HS and cell body secondary receptor binding.

Next, we investigated the onset of PsV translocation after CytD wash off. Cells were treated with PsVs and CytD for 5 h, washed, and incubated further without PsVs/CytD for 0 min, 30 min or 60 min, followed by fixation and staining for PsVs (Figure 2A [↗](#), magenta LUT) and F-actin (Figure 2A [↗](#), green LUT). The antibody staining against PsVs, like above the click-chemistry staining, is again highly variable. For analyzing the translocation process, we calculated the Pearson correlation coefficient (PCC) between PsV-L1 and F-actin staining. The PCC quantifies the similarity between two variables, in this case the pixel values of the two images. The PCC results in the value of 1 if images overlap perfectly and -1 for an image and its negative. After 5 h of CytD, PsVs again accumulate at the edge of the F-actin staining (Figure 2A [↗](#), compare to Figure 1B [↗](#) that shows membrane sheets). The distal PsVs and the F-actin staining partially exclude each other, thus at 0 min we obtain a negative PCC (Figure 2A and D [↗](#)). After 30 min and 60 min, PsVs mostly overlap with F-actin stained areas, which results in positive PCCs (Figure 2A and D [↗](#)). The time course of the change in PCC values suggests an onset of translocation recovery within 30 min after CytD removal.

Assessing previously proposed steps in HPV infection in our assay

PsVs reportedly bind tightly to their primary attachment site until the capsid surface undergoes structural changes, which involves modifications of L1 and L2 (see above). Consequently, we would expect that inhibition of L1 processing during the CytD incubation prevents the recovery of PsV translocation from the ECM to the cell body (Figure 2A and D [↗](#)). To test for this possibility, as employed in earlier studies, the protease inhibitor leupeptin was used to inhibit proteases including KLK8 which is required for L1 cleavage (Cerqueira et al., 2015 [↗](#)). Employing this inhibitor, the PCC between PsV-L1 and F-actin staining remains negative after CytD removal, showing that for translocation indeed the action of proteases is required (Figure 2B and D [↗](#)). In contrast, inhibition of L2 cleavage by a furin specific inhibitor has no effect on the PCC (Figure 2C and D [↗](#)). However, it should be noted that we occasionally observe PsVs not completely translocating but accumulating at the border of the F-actin stained area (for example see Figure 2C [↗](#) (60 min)). This results in an increase of the PCC almost equal to complete translocation, explaining why the PCC remains unaffected despite a furin inhibitory effect. Hence, furin inhibition may have some effect on translocation that, however, is undetected in this type of analysis.

As outlined above, during the 5 h incubation with CytD, proteases in the ECM are expected to cleave HS chains. These cleavage products should be able to diffuse out of the ECM, unless they remain associated with non-translocating PsVs. On the other hand, in the control, less HS cleavage products are trapped and if trapped the PsV leave the ECM. So far, and throughout the entire study, we stain the cells without prior cell permeabilization (albeit fixation perforates the cell membrane to some extent) in order to monitor only cell surface events. Using an antibody that reacts with an epitope in native heparan sulfate chains, only after CytD and if PsVs were present, the level of HS staining was significantly increased (Figure 3B [↗](#)). As shown in Figure 3A [↗](#), stronger HS staining at PsVs (open arrows) and as well in PsV free areas (closed arrows) was observed.

As additional control and shown in Supplementary Figure 2 [↗](#), we used an antibody that reacts with a HS neo-epitope generated by heparitinase-treated heparan sulfate chains (Yokoyama et al., 1999 [↗](#); for details see methods). This neo-epitope staining is independent of the presence of CytD and the incubation time, suggesting that CytD does not directly affect HS processing. Collectively, our findings indicate that without actin-dependent PsV translocation HS cleavage products are retained in the ECM, consistent with the hypothesis that cleaved HS remains associated with PsVs (Ozbun and Campos, 2021 [↗](#)).

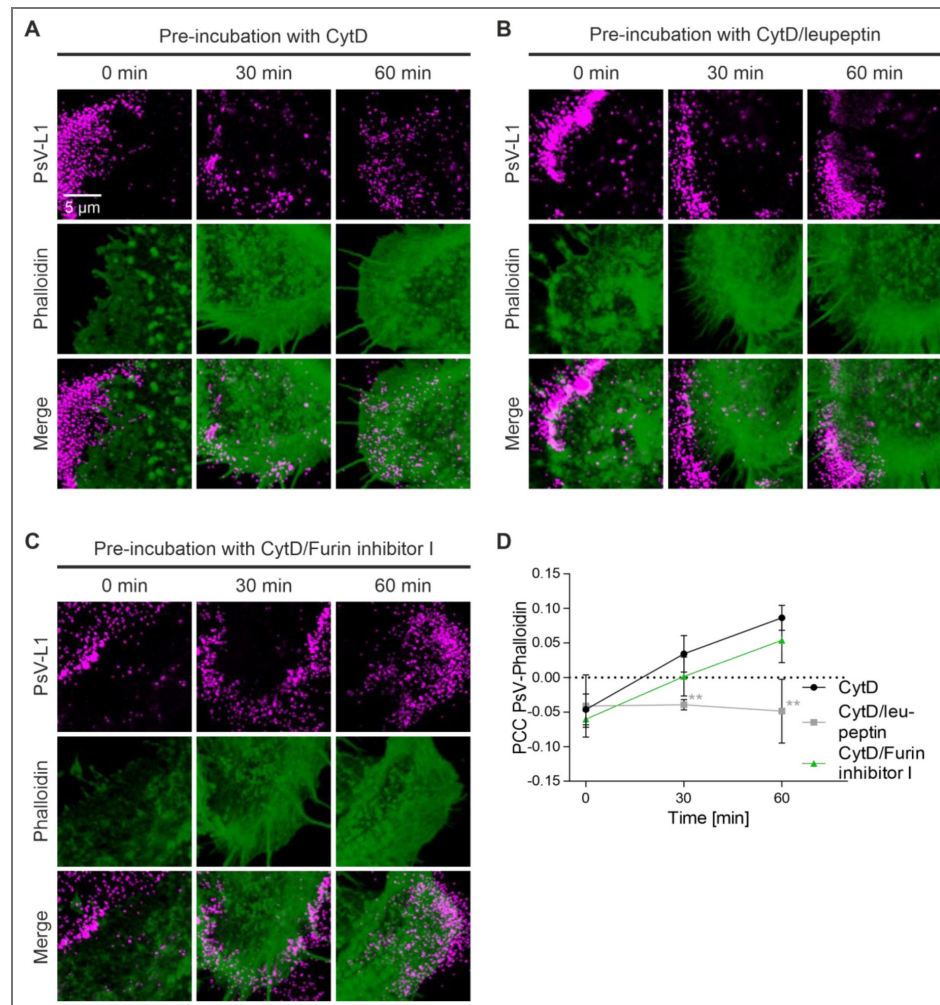


Figure 2. Transfer of PsVs from the ECM to the cell body requires PsV priming.

HaCaT cells were pre-incubated for 5 h at 37 °C with PsVs, in the presence of (A) 10 µg/ml CytD (CytD), (B) 10 µg/ml CytD and 100 µM leupeptin (CytD/leupeptin), or (C) 10 µg/ml CytD and 5 µM Furin inhibitor I (CytD/Furin inhibitor I). Afterwards, cells were washed and incubated without PsVs/inhibitors further for 0 min, 30 min or 60 min, before they were fixed and stained by indirect immunofluorescence for L1 (STAR GREEN, magenta LUT) and for F-actin by iFluor647-labelled phalloidin (green LUT). PsVs-L1 and F-actin staining were imaged in the confocal mode of a STED microscope. (D) For determination of the Pearson correlation coefficient (PCC) between PsV-L1 (magenta LUT) and Phalloidin (green LUT), we placed large ROIs onto the images that covered mainly the cell body but included parts of the cell periphery as well. The PCC was plotted over time. Values are given as means ± SD (n = 3 biological replicates). Statistical difference between CytD and CytD/inhibitors was analyzed by using the two-tailed, unpaired student's *t*-test (n = 3, for details see methods).

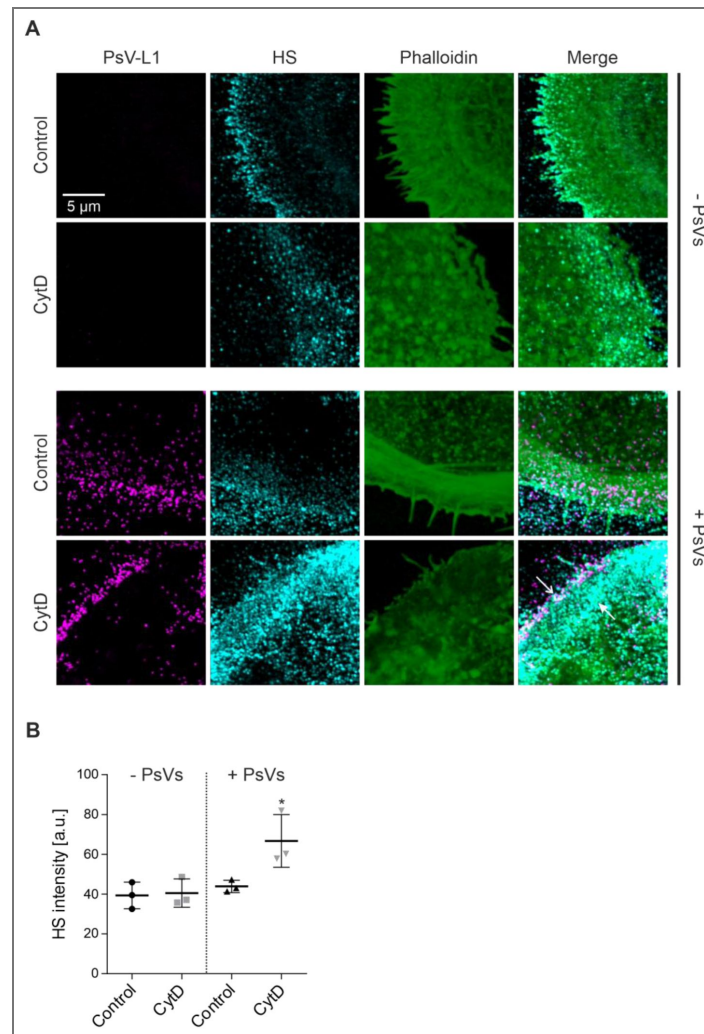


Figure 3. Increased HS intensity after incubation with PsVs and CytD.

(A) HaCaT cells were incubated without (top) and with (bottom) PsVs at 37 °C for 5 h, in the absence (Control, upper panels) or presence of 10 µg/ml CytD (CytD, lower panels). Afterwards, cells were washed, fixed and stained. Immunofluorescence was used for L1 (STAR GREEN; magenta LUT) and for HS (AlexaFluor 594; cyan LUT). F-actin was stained by iFluor647-labelled phalloidin (green LUT). PsVs-L1, HS and F-actin staining were imaged in the confocal mode of a STED microscope. The open arrow marks a region where PsVs overlap with HS. The closed arrow marks a region devoid of PsVs showing strong HS staining. (B) For analysis of the mean HS intensity, we placed large ROIs onto the images covering mainly the cell body but including parts of the cell periphery as well. Values are given as means $\pm$ SD ($n = 3$ biological replicates). Statistical differences between Control and CytD were analyzed by using the two-tailed, unpaired student's *t*-test ($n = 3$, for details see methods).

The transfer of PsVs from the ECM to the cell body is fast and efficient

Next, we studied the time course of PsV translocation in more detail. Cells were treated as in Figure 1A (in Figure 4, the 0 min time point is identical), followed by removal of PsVs/CytD. Different from Figure 2, we did not calculate the PCC that provides only an approximate idea of the time course of translocation, but monitored the diminishment of the integrated density of PsVs locating at the cell periphery after 0, 15, 30 and 60 min. The cell periphery is delineated from the cell body by the TMA-DPH membrane staining (Figure 4A, the cell periphery is defined by the area enclosed by the two white lines in the images shown with the magenta LUT).

At 0 min, compared to the control, CytD causes a 6-fold increase of PsV signal in the periphery (Figure 4B). This increase is more than halved when cells were incubated for 15 min, and after 30 min, the level of the control is reached (Figure 4B). This suggests that the half-time of PsV translocation from the periphery to the cell body is about 15 min. In fact, the half-time maybe longer, as we cannot exclude that cell spreading after CytD removal contributes to less PsVs measured in the cell periphery. In any case, translocation is fast and efficient and not a major component in the time course of infection. We conclude that translocation is not responsible for the asynchronous virion uptake observed for HPV PsVs.

Translocation of PsVs to CD151

During translocation, at some point PsVs may approach to cell surface HPV entry factors, like the tetraspanin CD151. For studying this possibility, we employ superresolution STED microscopy, analyzing the association of PsVs with CD151 over time. Cells are treated and monitored as above, but with an extended time window of up to 180 min, as cell surface processes are expected to take longer than an hour.

After fixation, PsVs and CD151 were double-stained with antibodies against L1 (Figure 5A, magenta LUT) and CD151 (Figure 5A, green LUT). In addition, we stained F-actin with fluorescent labelled phalloidin. We simultaneously image L1 and F-actin in the confocal and CD151 in the STED channel.

As shown in Figure 5A, CD151 concentrates in spots scattered across the cell surface. It is also present at cell protrusions that are rich in F-actin and vary strongly in number and shape (Supplementary Figure 3A shows some examples of these filopodia). Compared to the control, CytD again preserves the accumulated PsVs at the cell periphery, and at early time points PsVs appear brighter (Figure 5A; for overview images and F-actin staining see Supplementary Figure 4A and B; for variability of PsV accumulations after CytD see Supplementary Figure 4B).

Initially, we wondered whether, after reaching the cell body, PsVs are transported quickly further towards the center of the cell body. Based on the CD151 image, a cell border region was defined (for details see Supplementary Figure 3B). We counted the number of PsVs in this region, and expressed it as percentage of all PsVs in the image. In the control, the fraction of PsVs in the cell border region was rather low, ranging from 13.4% (0 min) to 7.6% (180 min). After CytD, at 0 min, compared to the control the fraction almost tripled (36.5%). It diminished to 16.9% after 60 min, and 12.3% after 180 min (Supplementary Figure 3C). This suggests that PsVs only briefly remain in the cell periphery.

The intensity of the CD151 staining between cells is highly variable (Figure 5A and B). In the control, the mean CD151 intensity shows no trend over the 180 min time course, whereas with CytD the intensity diminishes after 180 min (Figure 5B). The decrease of the CD151 staining intensity points towards the possibility that, after CytD wash off, CD151 is more strongly internalized, presumably due to co-internalization with endocytosed PsVs. This idea is supported by the observation that, in particular at CytD/180 min, we occasionally observe CD151/PsV agglomerations (Figure 5A, see structures marked by arrows at CytD/180 min, for more examples see Supplementary Figure 5A). We did not study this issue systematically, but some of these structures have clear three-dimensional extension (see Supplementary Figure 5B for axial

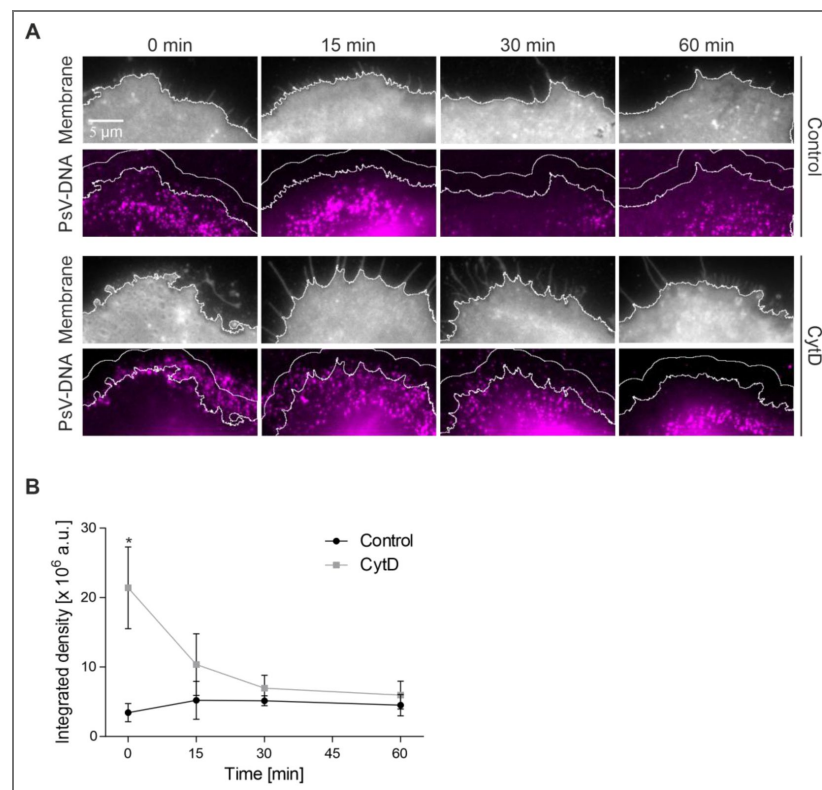


Figure 4. Fast diminishment of accumulated PsVs at the cell periphery after removal of CytD.

(A) HaCaT cells were incubated with PsVs at 37 °C for 5 h, in the absence (Control) or presence of 10 μ g/ml CytD (CytD). Then, cells were washed and incubated for the indicated time periods without PsVs/CytD, before they were fixed and stained as in Figure 1 (t = 0 min is identical to Figure 1A; for clarity we show only the membrane (gray LUT; images are shown at different settings of brightness and contrast) and the PsV-DNA staining (magenta LUT; images are shown at the same settings of brightness and contrast). The white lines in the membrane images delineate the cell body from the periphery. They were created with reference to the membrane staining (for details see methods). Using the cell body delineation as starting point, an up to 30-pixel broad area was created (PsV channel, magenta LUT; see area enclosed by the smoother white lines and the cell body delineation lines). The area enclosed by the two white lines defines the cell periphery. (B) The PsV-DNA signal of the periphery was quantified as integrated density, background corrected, and plotted over time. Values are given as means $\pm$ SD (n = 3 biological replicates). The statistical difference between the same time points of Control and CytD were analyzed by using the two-tailed, unpaired student's *t*-test (n = 3, for details see methods).

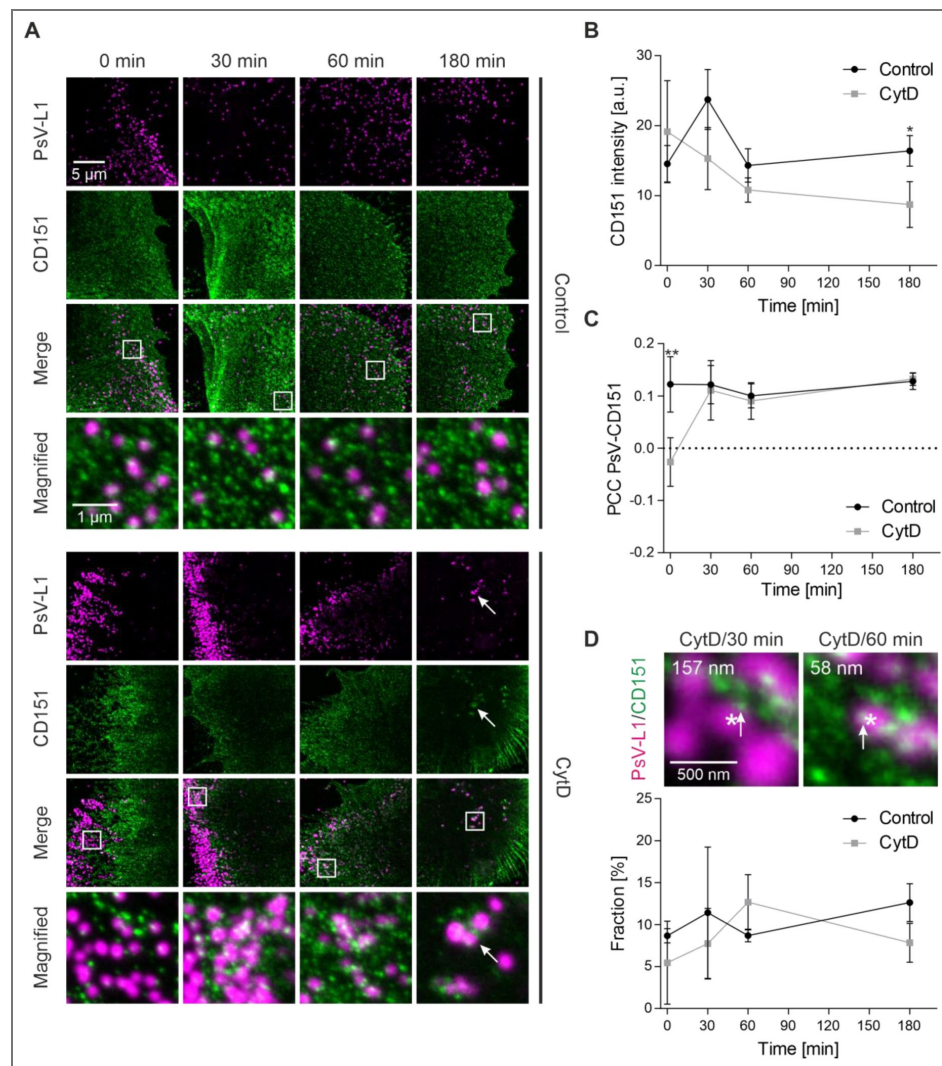


Figure 5. Association between PsVs and CD151 is an early event in the infection cascade.

(A) HaCaT cells were incubated with PsVs at 37 °C for 5 h, in the absence (Control, upper panels) or presence of 10 µg/ml CytD (CytD, lower panels). Afterwards, cells were washed and incubated without PsVs/CytD further for 0 min, 30 min, 60 min or 180 min, before they were fixed and stained by indirect immunofluorescence for L1 (STAR GREEN; magenta LUT) and for CD151 (AlexaFluor 594; green LUT), and for F-actin by iFluor647-labelled phalloidin (here not shown for clarity, please see Supplementary Figure 3 for F-actin illustrating the variability of filopodia after CytD treatment). The bottom rows show magnified views of the merged images from the regions marked by the white boxes. PsV-L1 and F-actin staining were imaged in the confocal and CD151 staining in the STED mode of a STED microscope, respectively. Therefore, compared to CD151, the PsVs are less resolved and also appear much larger than their real physical size (see magnified views). CytD/180 min, arrows mark presumably endocytic structures that formed in the central cell body region (for more examples see Supplementary Figure 5). For analysis, we placed large ROIs onto the images that covered mainly the cell body but included parts of the cell periphery as well (for an example ROI see Supplementary Figure 6A). (B) The mean CD151 intensity was measured and plotted over time. (C) The PCC between PsV-L1 (magenta LUT) and CD151 (green LUT) was calculated and plotted over time. (D) The fraction of PsVs (in percent) that have a distance to the next neighbored CD151 maximum ≤ 80 nm, which we define as closely associated, is plotted over time. Please note that the values in (D) were corrected for random background association (for details see Supplementary Figure 7). Two examples of PsVs (each marked by an asterisk) from the CytD/30 min (left) and CytD/60 min (right) conditions are shown. The value in the upper left states the shortest distance between the PsV and the next nearest CD151 maximum (marked by an arrow) in nm. Values are given as means $\pm$ SD ($n = 3$ biological replicates). Statistical differences between the same time points of Control and CytD were analyzed by using the two-tailed, unpaired student's *t*-test ($n = 3$, for details see methods).

scans). Therefore, they are likely tubular structures filled with several PsVs, as previously described by electron microscopy (Schelhaas et al., 2012). We observe fewer of such structures at control/180 min, probably because cells have been actively interacting with PsVs for altogether 8 h, opposed to 3 h in the CytD treated cells. Hence, in the control, PsV endocytosis may be less synchronized compared to CytD explaining the CD151 diminishment at CytD/180 min (Figure 5B).

For studying the association between PsVs (L1) and CD151, the PCC between the channels is calculated. The PCCs are around 0.1, with the exception of the CytD/0 min value that is significantly lower and even negative (Figure 5C, for PCC values of flipped images see Supplementary Figure 6). This reflects the somewhat mutual exclusion of the two stainings as already discussed above; PsV accumulations are at the cell periphery and the CD151 staining is mainly at the cell body and in filopodia. At 30 min after CytD, the PCC has reached the level of 0.1 as in the other conditions, suggesting that actin processes are required early for the PsV-CD151 association.

One would expect that the translocation of PsVs to the cell surface increases the number of PsVs that are closely associated with CD151. As criteria for close association, we define a distance of ≤ 80 nm between PsV and CD151 maxima, a value which is close to the resolution limit of the used microscope (Finke et al., 2020). In the control, the fraction of PsVs closely associated with CD151, after correction for random background association (see Supplementary Figure 7), is 8.7% (Figure 5D, control). At 0 min after CytD, we start with a fraction of 5.5%, a value that increases to 12.7% in the next 60 min (CytD/60 min), although this change is not significant (Figure 5D, CytD).

Altogether, we conclude that after a 5 h pre-incubation with PsVs (Figure 5D, control), about 10% of the PsVs are associated closely with CD151 in a time window of 180 min. CytD diminishes only the early association at 0 min (5.5%, Figure 5D, CytD). The data suggests that PsVs establish contact to CD151 assemblies early in the infection cascade (see also the increase in the PCC after CytD between 0 and 30 min, Figure 5C).

PsV association with HS

Next, we studied the association between PsVs and HS. As a reference staining for the cell body, we used Itga6 that is not visible at cell protrusions. PsVs are visualized by click-chemistry and imaged in the confocal channel. HS and Itga6 were stained by antibodies, and imaged at STED microscopic resolution. The three stainings were simultaneously recorded.

As shown in Figure 6A (green LUT), the Itga6 staining results in a pattern of dense spots. The Itga6 intensity does not change over time (Supplementary Figure 8E). The pattern of the HS staining (cyan LUT) and the overlap of HS with PsVs and Itga6 are highly variable (Figure 6A).

CytD increases the intensity of HS (Figure 6B; also apparent when comparing the HS brightness in the upper and lower panels of Figure 6A; for an overview of the CytD/0min images see Supplementary Figure 4C). This increase of intensity is particularly notable at the 0 min time point, where the samples treated with CytD have a more than two-fold higher intensity and differ significantly from the control. Hence, in this experiment we reproduce the PsV/CytD mediated HS intensity increase observed in Figure 3.

In the control, the PCC between PsVs and HS is little above zero. The largest PCC is found at CytD/0 min, which reflects the finding that at this time point both PsVs and HS preferentially locate at the cell periphery (Figure 6C, for PCC values of flipped images see Supplementary Figure 8). Over time, the accumulated PsVs diminish due to translocation, which is accompanied by a PCC approaching zero, the same value as in the control (Figure 6C). Additionally, we analyze the PCC between PsVs and HS specifically in the cell body region, excluding the cell periphery (Figure 6D). For the control, the PCC in the cell body region ranges between 0 and 0.05. After CytD, we observe an increase in the PCC from 0 min to 30 min and a decrease from 30 min to 180 min (for PCC values of flipped images see Supplementary Figure 8).

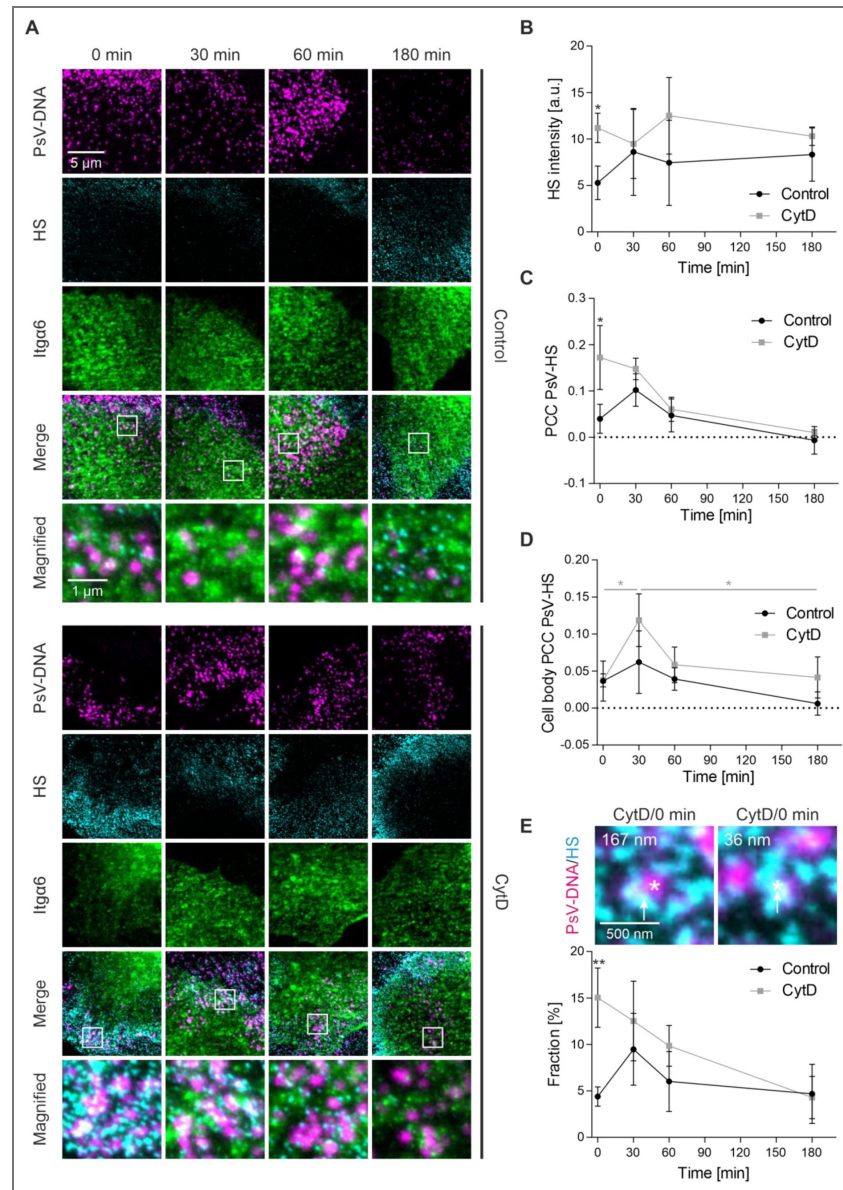


Figure 6. Association between PsVs and HS.

(A) HaCaT cells were incubated with PsVs at 37 °C for 5 h, in the absence (Control, upper panels) or presence of 10 μ g/ml CytD (CytD, lower panels). Afterwards, cells were washed and incubated without PsVs/CytD further for up to 180 min, before they were fixed and stained. PsVs (magenta LUT) were visualized by click-chemistry (6-FAM Azide) and indirect immunolabeling was used for HS (AlexaFluor 594; cyan LUT) and for Itga6 (STAR RED; green LUT). The bottom rows show magnified views of the white boxes in the merged images. PsV-DNA staining was imaged in the confocal and HS and Itga6 staining in the STED mode of a STED microscope. For analysis, we placed large ROIs onto the images that covered mainly the cell body but included parts of the cell periphery as well (see example in Supplementary Figure 8A). For (D), smaller ROIs covering only the cell body region were used. (B) The mean HS intensity plotted over time. (C) The PCC between PsV-DNA (magenta LUT) and HS (cyan LUT) over time. (D) The PCC between PsV-DNA (magenta LUT) and HS (cyan LUT) in the region of the cell body over time. (E) The fraction of PsVs (in percent) closely associating with HS (distance $\leq$ 80 nm) plotted over time (for background correction see Supplementary Figure 9). Two examples of PsVs (each marked by an asterisk) from the CytD/0 min condition are shown. The value in the upper left corner states the shortest distance (in nm) between the marked PsV and its next nearest HS maximum (marked by an arrow). Values are given as means $\pm$ SD ($n = 3$ biological replicates). Using the two-tailed, unpaired student's t -test ($n = 3$ biological replicates), we analyzed in (B), (C) and (E) the statistical differences between the same time points of Control and CytD, and in (D) the difference between CytD/30 min and CytD/0 min or CytD/180 min (for details see methods).

When analyzing the PsVs closely associating with HS, at 0 min, CytD increases the fraction of PsVs associated with HS more than 3-fold. Over the next 180 min, this fraction gradually decreases until it is equal to the control value (Figure 6E [↗](#)).

Altogether, the analysis of the PCC between HS and PsVs (Figure 6C [↗](#)) along with the fraction of closely HS associated PsVs (Figure 6E [↗](#)) indicates that CytD treatment increases HS/PsV association at 0 min, likely because PsVs remain accumulated in the ECM. After CytD removal, this association diminishes. A transient increase in PCC is observed at 30 min specifically in the region of the cell body (Figure 6D [↗](#)) which likely reflects the translocation of HS-coated virions to the cell body. However, this association also diminishes over the subsequent 150 min, suggesting that an increasing number of PsVs are no longer associated with HS – likely due to progressive loss of the HS coat over time. Alternatively, PsVs just may disappear after internalization.

Next, for each PsV we plot the ‘distance to the next nearest Itga6 maximum’ against ‘the distance to the next nearest HS maximum’ (Figure 7 [↗](#)). In the control, the distances remain unchanged over the entire observation time; 59% - 65% of the PsVs are at a short distance (< 250 nm) to both Itga6 and HS (Figure 7B [↗](#), black; see also Supplementary Table 1 [↗](#)), 12 - 15% are at a short distance (< 250 nm) to HS but not to Itga6 (> 250 nm) (Figure 7B [↗](#), green; see also dashed green box in Figure 9 [↗](#) marking PsVs in the ECM). Also, the distance plots are in line with the idea that the distance patterns of PsVs in untreated cells do not change over the 180 min observation time (Figure 7C [↗](#), please compare the upper distance plots from left to right). In cells treated with CytD, a larger fraction of PsVs is accumulated at the periphery at the 0 min time point, which is reflected in a larger fraction (46% as opposed to 12% in the control; Figure 7B [↗](#), green; Supplementary Table 1 [↗](#)) of PsVs with a short distance (< 250 nm) to HS but not to Itga6. Over time, the PsVs in the CytD treated cells acquire a shorter distance to Itga6 and a larger distance to HS (Figure 7C [↗](#), lower row). The population with short distances to HS and large ones to Itga6 strongly diminishes from 46% to 11% (Figure 7B [↗](#), CytD, green). After 180 min, the distances are similar to the untreated control. Between CytD/60 min and CytD/180 min, the fraction of PsVs with a large distance (> 250 nm) to HS and a short distance (< 250 nm) to Itga6, representing PsVs at the cell body without HS (see also dashed magenta box in Figure 9 [↗](#)), increases from 9.6% to 25.2% (Figure 7B [↗](#), magenta). This is inconsistent with the idea of endocytosis of HS-coated PsVs that would only diminish short distances to Itga6 but not create long distances to HS, which is what we observe. The observation supports the idea that PsVs lose their HS coat after translocating to the cell surface, which is also supported by the transient increase in the PCC between HS and PsVs of the cell body at 30 min (Figure 6D [↗](#)).

The movement of HS towards the cell body after removal of CytD, which indirectly demonstrates that PsVs are coated with HS, is suggested by a shortening of the HS-Itga6 distance over time (Supplementary Figure 8D [↗](#)). Moreover, the PCC between PsVs and HS in the cell body area increases (Figure 6D [↗](#)). Altogether, the data suggest that HS-coated PsVs migrate towards the cell body. A large fraction of translocating PsVs sheds HS within one hour after removing CytD (Figure 7C [↗](#), lower panels, showing increasing PsV-HS distances). This is the time window where we notice the occurrence of endocytic PsV/CD151 structures (see above and Supplementary Figure 5 [↗](#)).

Actin retrograde transport, which underlies the here observed virion transport, is the integrative result of three components (Schelhaas et al., 2008 [↗](#); Smith et al., 2008 [↗](#)). On the one side, actin filaments in the cell periphery are pushed into the cell body area when actin polymerizes at their tips. The retrograde filament movement causes tension and is facilitated by F-actin degradation opposed to the side of F-actin growth. Moreover, the motor protein myosin II exerts force on actin filaments, pulling those towards the cell body. As CytD broadly interferes with F-actin dependent processes, we investigated the effects upon inhibition of only one of the three components, namely the myosin II mediated retrograde movement towards the cell body. Instead of CytD, we employed in the 5 h preincubation the myosin II inhibitor blebbistatin. For the control (0 min), we show in Figure 8A [↗](#) one example of a cell with comparatively many PsVs at the periphery (as mentioned above, the PsV pattern is highly variable) to better illustrate the difference to the PsV pattern occasionally seen with blebbistatin. After blebbistatin treatment (0 min), PsVs are still distal to the

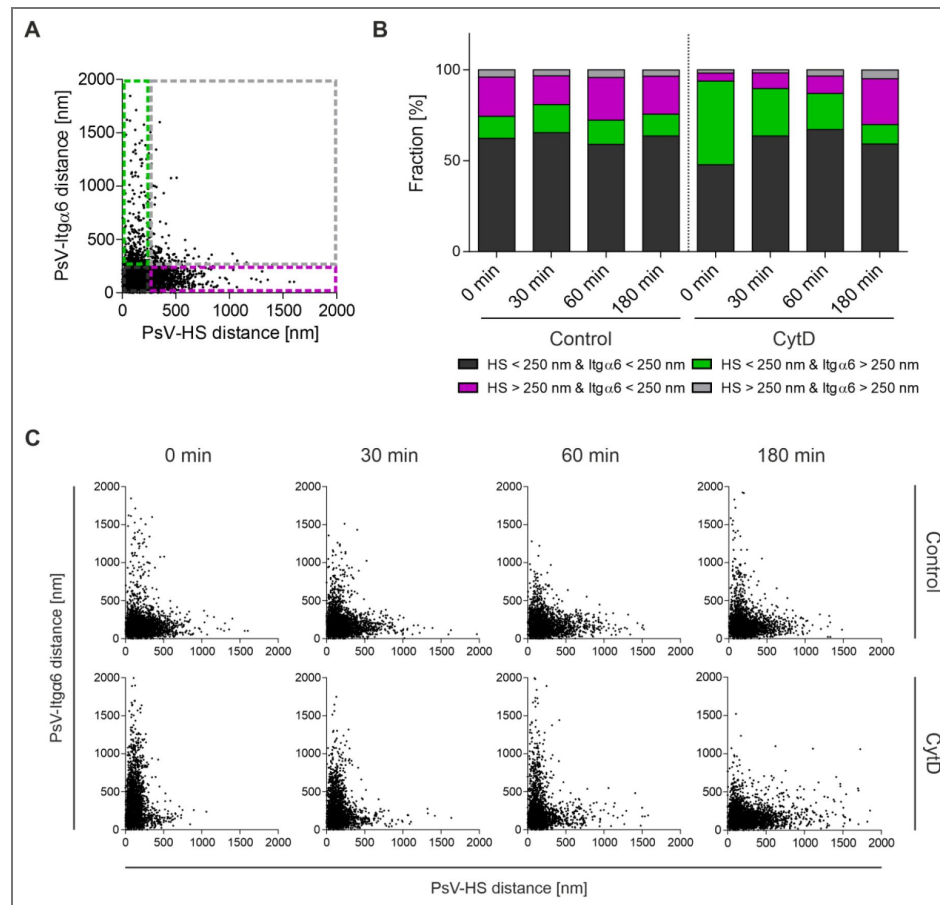


Figure 7. PsV-Itga6 and PsV-HS distances over time.

(A) Definition of four PsV populations based on the PsV distances to Itga6 and HS (the plot is taken from Control, 0 min and shown again in C). Dashed green rectangle, PsVs with a distance to HS < 250 nm and to Itga6 > 250 nm. Dashed magenta rectangle, PsVs with a distance to HS > 250 nm and to Itga6 < 250 nm. Dashed gray square, PsVs with a distance to HS > 250 nm and to Itga6 > 250 nm. PsVs not included in the previous categories have a distance to HS < 250 nm and to Itga6 < 250 nm (dashed black square). (B) From the PsVs analyzed in Figure 6, for the Control (left) and CytD (right) the PsV fraction size (in percent) of each population is illustrated. Shown are the means of three biological replicates. For means $\pm$ SD and statistical analysis see Supplementary Table 1. (C) For the Control (top) and CytD (bottom), for each PsV, we plotted the shortest distance to Itga6 against the shortest distance to HS (pooling the three biological replicates; 3,043 – 4,080 PsVs per plot).

cell body but less dispersed than after CytD treatment, seemingly as if translocation started but stopped in the midst of the pathway (Figure 8A [↗](#), blebbistatin). The PCC between PsVs and HS, like after CytD (Figure 6C [↗](#)), is elevated after blebbistatin, albeit the effect is not significant (Figure 8C [↗](#)). The cell body PCC, is not at 30 min (CytD) but already at 0 min elevated (compare Figure 6D [↗](#) to Figure 8D [↗](#)), which can be explained by partial translocation. This is further supported by the fact that only 8% of PsVs are closely associated with HS (Figure 8E [↗](#); blebbistatin, 0 min) compared to 15% after CytD treatment (Figure 6E [↗](#); 0 min).

Furthermore, after 0 min PsV incubation with blebbistatin we observe no effect on the HS intensity (compare Figure 8B [↗](#) to Figure 3B [↗](#) and Figure 6B [↗](#)). Hence, in contrast to CytD, blebbistatin does not trap the PsVs in the ECM where they associate with HS, but ongoing actin polymerization pushes actin filaments along with PsVs towards the cell body.

Discussion

In this study, we investigate the early events of HPV16 infection preceding virus endocytosis. Our findings reveal that actin dynamics of filopodia, rather than passive virion diffusion, are crucial for releasing HS-coated virus particles from the adhesive ECM and translocating them to the cell body. When actin-dependent processes are inhibited by CytD, HS-coated virus particles remain trapped in the ECM, allowing for a more synchronized observation of the infection process. Once the transfer is no longer blocked, PsVs rapidly migrate toward the cell body, associate with CD151, and shed their HS-coat before endocytosis.

A reversible blocking of PsV translocation from the ECM to the cell body by CytD

Many viruses use cell surface HS for primary attachment, including herpes simplex virus type 1, human cytomegalovirus, human immunodeficiency virus type 1, adenovirus type 2, dengue virus, hepatitis B virus, and vaccinia virus (Cagno et al., 2019 [↗](#); Giroglou et al., 2001 [↗](#); Tian et al., 2021 [↗](#)). HPVs could be different as they bind as well to HS of the extracellular basement membrane, at least *in vivo* during the wounding and healing processes required for infection (Day and Schelhaas, 2014 [↗](#); Kines et al., 2009 [↗](#); Ozbun and Campos, 2021 [↗](#); Schiller et al., 2010 [↗](#)).

By treating cells with CytD, we block actin-mediated transport, leading to a 6-fold accumulation of PsVs at the cell periphery after 5 hours (Figure 4B [↗](#)). After CytD removal, infectivity is only moderately reduced (29%) (Figure 1E [↗](#)), suggesting that the treatment is reversible and only slightly harmful, if at all.

Upon CytD washout, PsVs translocate to the cell body with a half-time of ≈ 15 minutes, as suggested by the diminishment of accumulated PsVs at the periphery. As mentioned above, the half-time could be longer if cell spreading is in part responsible for the translocation of PsVs onto the cell body. However, we assume that this is rather unlikely, as cell spreading would increase the PCC between PsVs and F-actin under a condition where filopodia mediated transport is blocked but not cell spreading, which is not the case (Figure 2B and D [↗](#), CytD/leupeptin). Within this 15 min time frame, PsVs associate more strongly with CD151. It is in principle possible that along with CytD removal ECM-accumulated PsVs are merely washed off. However, after only 15 min, the effect of this could not have been very strong, considering the still 6-fold accumulation of PsVs after the 5 h incubation with CytD. Furthermore, CytD pre-treatment appears to synchronize uptake: We observe endocytic structures after CytD treatment more frequently (for examples see Supplementary Figure 5 [↗](#)) and the CD151 intensity diminishes over time only after the CytD treatment (Figure 5B [↗](#)). We therefore conclude that the processing of the PsVs in the ECM is contributing to desynchronization and is largely completed after the 5 h pre-incubation with CytD. We estimate that about 6-fold more PsVs (Figure 2B [↗](#)) translocate to the cell surface in a coordinated fashion after CytD removal, which likely enhances the number of endocytic events compared to untreated controls.

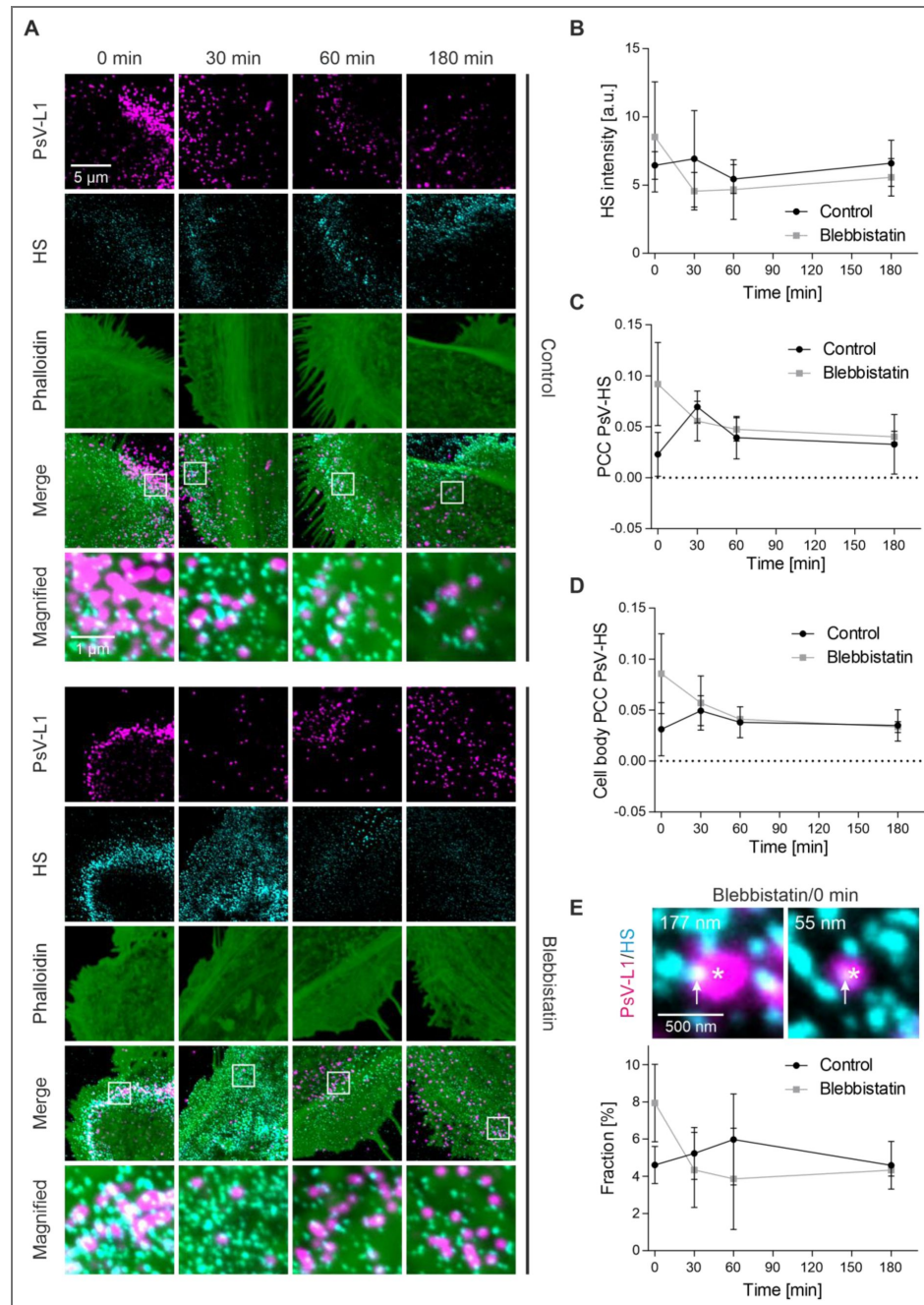


Figure 8. Association between PsVs and HS after blebbistatin treatment.

(A) HaCaT cells were incubated with PsVs at 37 °C for 5 h, in the absence (Control, upper panels) or presence of 30 μ M blebbistatin (Blebbistatin, lower panels). Afterwards, cells were washed and incubated without PsVs/blebbistatin further for up to 180 min, before they were fixed and stained. Immunofluorescence was used for L1 (STAR GREEN; magenta LUT) and HS (AlexaFluor 594; cyan LUT) staining. F-actin was stained by iFluor647-labelled phalloidin (green LUT). The bottom rows show magnified views of the white boxes in the merged images. PsVs and F-actin staining were imaged in the confocal and HS staining in the STED mode of a STED microscope. For analysis, we placed large ROIs onto the images that covered mainly the cell body but included parts of the cell periphery as well. For (D), smaller ROIs covering only the cell body region were used. (B) Mean HS intensity over time. (C) PCC between PsV-L1 (magenta LUT) and HS (cyan LUT) over time. (D) PCC between PsV-L1 (magenta LUT) and HS (cyan LUT) in the region of the cell body over time. (E) The fraction of PsVs (in percent) closely associating with HS (distance $\leq$ 80 nm) plotted over time (for background correction see Supplementary Figure 11). Two examples of PsVs (each marked by an asterisk) from the blebbistatin/0 min condition are shown. The value in the upper left states the shortest distance between the PsV and its next nearest HS maximum (marked by an arrow) in nm. Values are given as means $\pm$ SD ($n = 3$ biological replicates). Statistical differences between the same time points of Control and Blebbistatin were analyzed by using the two-tailed, unpaired student's t -test ($n = 3$), but the analysis yielded no p -values below 0.05.

The role of actin-driven virion transport

The strong electrostatic interaction between HPV and HS implies that some processing — via HS cleavage or structural capsid changes — is a prerequisite for the release of virions from the ECM. CytD likely does not inhibit HS processing (Jamieson et al., 2021 [↗](#); see also [Supplementary Figure 2](#) [↗](#)) nor is it likely to inhibit KLK8. Therefore, we assume that PsVs are primed after 5 h of incubation with CytD. They may remain associated via weaker interactions, and actin-driven transport is required for pulling the HS-coated virions out of the sticky matrix towards the cell body. Sticky PsV aggregations that are released from HeLa cell surface after binding and blocking of the subsequent entry pathway have been observed earlier and might reflect similar HS/PsV-structures (Mikulić et al., 2025 [↗](#)). Alternatively, an ECM bound virion is not exposed to all processing factors at the same time. In this case, migration through the ECM is additionally required to establish, one by one, contact with all processing enzymes involved. Once the fully processed virions have reached the cell surface, detachment from the ECM may be facilitated by new bonds established between the PsVs and cell surface receptors. Now, the PsVs and cell surface receptors move towards the central cell body by intracellular dynamics which results in CD151 accumulation at virus binding sites and ultimately endocytosis (see [Figure 5](#) [↗](#) as well as [Supplementary Figure 5](#) [↗](#)).

Both CytD and blebbistatin increase PsV-HS colocalization, but blebbistatin allows partial translocation (compare [Figure 6C](#) [↗](#) and [Figure 8C](#) [↗](#) as well as [Figure 6D](#) [↗](#) and [Figure 8D](#) [↗](#)), indicating that actin retrograde flow initiates movement, while myosin II contributes to sustained transport. As a result, virus movement can proceed to some extent after blebbistatin treatment. However, the subsequent blocking of transport at later time points reflects the inhibition of myosin II, which eventually disrupts the cytoskeletal dynamics and force generation that is necessary for sustained transport. In contrast, CytD directly interferes with actin polymerization, leading to an earlier and more persistent blocking of virus transport due to structural disruption of the filopodial tracks themselves.

Altogether, our findings support earlier reports showing that HPV utilizes filopodia for cell entry, migrating at several micrometers per minute (Schelhaas et al., 2008 [↗](#); Smith et al., 2008 [↗](#)). This translocation is fast compared to the entire infection process and therefore cannot largely contribute to the asynchronous HPV uptake. Since we use polarized cells, PsVs hardly bind to the cell surface directly ([Supplementary Figure 1](#) [↗](#)). Therefore, the actin-driven virion transport plays a decisive role in HPV infection of isolated HaCaT cells, which cannot be modelled in nonpolarized cells, where PsVs can bind to the entire cell surface.

HaCaT cells as model system in HPV infection

Inhibition of HPV16 PsV transport diminishes infection only of subconfluent but not of confluent HaCaT cells by about 50% (Schelhaas et al., 2008 [↗](#)). Therefore, actin-dependent translocation along protrusions may be dispensable for infection at a high cell density (Schelhaas et al., 2008 [↗](#)) and merely increases the exposure of cells to virions (Smith et al., 2008 [↗](#)). However, this increased exposure could be relevant *in vivo*, as wounding of the epidermis results in upregulation of filopodia formation (Vasjoukhin et al., 2000 [↗](#)). Filopodia usage not only facilitates infection but also increases the likelihood of virions to reach their target cells during wound healing, namely the filopodia-rich basal dividing cells. In fact, several types of viruses exploit filopodia during virus entry (Chang et al., 2016 [↗](#)), hinting at the possibility that for HPV and other types of viruses actin-driven virion transport may play a more important role than it is currently assumed. If this is the case, sub-confluent HaCaT cells, or even better single HaCaT cells, would be an ideal model system for the study of these very early infection steps that involve ECM attachment and subsequent filopodia-dependent transport. As shown in [Supplementary Figure 1](#) [↗](#), HaCaT cells have many binding sites for the HPV16 PsVs. However, as they are polarized and the binding receptors are only at the basal membrane, they remain relatively inaccessible by diffusion. Therefore, the ECM binding that is also observed *in vivo* (Day and Schelhaas, 2014 [↗](#)) and subsequent transport via filopodia are used upon infection of HaCaT cells that locate at the periphery of cell patches. Here, PsVs bind to the ECM which strongly enhances infection of primary keratinocytes (Bienkowska-

Haba et al., 2018 [\[4\]](#)). In contrast, HPV can readily bind to HSPGs on the cell surface of nonpolarized cells, and by this bypasses ECM mediated virus priming and the filopodia dependency. We propose that HaCaT cells are a valuable system for studying the very early events in HPV infection that allows for dissecting capsid interaction with ECM resident priming factors and cell surface receptors.

Translocating PsVs are decorated with HS which they lose over time

CytD treatment increases the HS signal when PsVs are present (Figure 3B [\[4\]](#) and 6B [\[4\]](#)). A relationship between PsVs and HS cleavage products has been previously suggested (Ozbun and Campos, 2021 [\[4\]](#); Surviladze et al., 2012 [\[4\]](#)), and we propose that our data is in line with these previous observations. It should be noted that under conditions without the 5 h CytD blocking, PsVs may only briefly reside in the ECM and as a result are less decorated with HS. Hence, under our assay conditions the HS decoration of virions may be enhanced in comparison to other assays.

At the 0 min time point post-CytD, 15% of the PsVs are closely associated with HS (Figure 6E [\[4\]](#)), but this fraction declines to 4.3% by 180 min. Considering the important role of HS as a primary attachment site, the fraction of 15% appears unexpectedly small. There are multiple possible explanations for underestimating the number of PsVs closely associated with HS. One contributing factor may be that accumulated PsVs are not well-resolved (see above), which results in an underestimation of HS associated PsVs in particular at CytD/0 min. Other explanations include that the HS antibody does not reach all HS epitopes in the dense matrix. Yet another explanation could be that PsV binding to HS and antibody binding to HS is sometimes mutually exclusive, a hypothesis which is supported by data showing that the HPV capsid possesses multiple HS binding sites (Dasgupta et al., 2011 [\[4\]](#); Richards et al., 2013 [\[4\]](#)). Together, our data suggests that PsVs initially translocate with HS attached and then shed it, potentially either upon receptor binding or further capsid rearrangement. These observations are in line with a previous study showing reduced HS colocalization during productive infection (Selinka et al., 2007 [\[4\]](#)).

The initial HS-coat may facilitate the formation of a complex with PsVs and cell surface receptors such as integrins or growth factor receptors (GFRs) (Ballut et al., 2013 [\[4\]](#); Pellegrini, 2001 [\[4\]](#)). This could facilitate receptor clustering and engagement as shown previously for coronavirus infection (Bugatti et al., 2025 [\[4\]](#); Clausen et al., 2020 [\[4\]](#); Zhang et al., 2023 [\[4\]](#)), as well as the binding of signaling molecules to the cell surface (Miladinova et al., 2022 [\[4\]](#)). Here, CD151 may support receptor clustering by binding to both integrins and GFRs (Hemler, 2005 [\[4\]](#); Mikulić et al., 2019 [\[4\]](#)). In addition, earlier studies suggested that cofactors like laminin-332 and growth factors further stabilize these interactions (Culp et al., 2006 [\[4\]](#); Mikulić et al., 2019 [\[4\]](#); Richards et al., 2014 [\[4\]](#); Surviladze et al., 2012 [\[4\]](#)). Furthermore, the HPV capsid has multiple binding sites for distinct functions (Richards et al., 2013 [\[4\]](#)) which supports the idea that HS fulfills several roles during viral entry — first by promoting capsid attachment, then inducing conformational changes, and finally by assisting in receptor engagement and clustering. However, HS cleavage alone is not sufficient for ECM release; active transport and capsid priming remain essential (see leupeptin effect in Figure 2B and D [\[4\]](#)). Diffusion may account for some virion movement, but filopodia-mediated transport dominates in this context.

PsV-CD151 association is an early contact at the cell surface

CD151 is a known mediator of HPV entry (Scheffer et al., 2013 [\[4\]](#); Spoden et al., 2008 [\[4\]](#)). Our data shows that about 10% of the PsVs are closely associated with CD151 at all post-translocation time points. This fraction is lower at 0 min post-CytD (Figure 5D [\[4\]](#), see also lower PCC in Figure 5C [\[4\]](#)), indicating that actin disruption impairs initial CD151 engagement. Hence, the data suggest that PsVs establish contact to CD151 assemblies shortly after translocation to the cell surface, and from then on, the fraction of closely associated PsVs remains constant. At first glance this implies that, after the initial formation of the PsV-CD151 assemblies, they do not change in nature. However, we observe agglomerated CD151 maxima (locally patched maxima) close to PsVs at later time points that likely are endocytic structures (see arrow in Figure 5A [\[4\]](#), Supplementary Figure 5 [\[4\]](#)). This

agglomeration process is not detected in our analysis as we only determine the next nearest CD151 assembly. Some further CD151 reorganization into larger entry platforms has been previously observed (Florin and Lang, 2018 [\[18\]](#)). This is in accordance with the observation that virions enter the cell in crowds with many of them occupying one endocytic organelle (Schelhaas et al., 2012 [\[19\]](#)). For the PsVs closely associated with CD151, one may have expected a larger fraction than 10%. Given that PsVs are not supposed to bind directly to CD151 and that CD151 is likely part of a larger membrane structure, like a tetraspanin enriched microdomain, our ≤ 80 nm distance criteria may be too short and by this we underestimate the association frequency. Altogether, we propose that an early contact between PsVs and the cell surface involves an assembly defined by CD151 (Figure 9 [\[20\]](#)).

Integration of our data into the HPV infection cascade

HPV infection is the result of several steps, starting with the initial binding of virions via electrostatic and polar interactions (Dasgupta et al., 2011 [\[21\]](#)) to the primary attachment site HS (Richards et al., 2013 [\[22\]](#)), which induces capsid modification (Cerqueira et al., 2015 [\[23\]](#); Feng et al., 2024 [\[24\]](#)) and HS cleavage (Surviladze et al., 2015 [\[25\]](#)), enabling the virion to be released from the ECM or the glycocalyx. Next, virions bind to the cell surface to a secondary receptor complex that forms over time, and become internalized via endocytosis, before they are trafficked to the nucleus (Mikulić et al., 2021 [\[26\]](#); Ozbun and Campos, 2021 [\[27\]](#)). Regarding the transition from the primary attachment site to cell surface binding, as already outlined in the introduction, two models are discussed. In one model, proteases cleave the capsid proteins. After priming, the capsids are structurally modified and the virion can dissociate from its HS attachment site. It has been suggested that capsid priming is mediated by KLK8 (Cerqueira et al., 2015 [\[23\]](#)) and furin (Richards et al., 2006 [\[28\]](#)). In our system, KLK8 inhibition blocks PsV transport, while furin inhibition has some effect that, however, cannot be detected in this analysis (Figure 2 [\[29\]](#)) suggesting furin engagement at later steps in the infection cascade. This is in line with earlier *in vitro* studies on the role of cell surface furin (Day et al., 2008 [\[30\]](#); Day and Schiller, 2009 [\[31\]](#); Surviladze et al., 2015 [\[25\]](#)). In any case, our results align with both models of ECM detachment: one involving HS cleavage (HS co-transfer) and another involving capsid modification (by e.g., KLK8).

We propose that after 5 h of CytD treatment, glycan-induced structural activation as well as capsid processing by proteases such as KLK8 of the capsid and HS cleavage essentially are completed ((i) in Figure 9 [\[20\]](#)). Subsequently, (ii) the HS decorated virion is transported from the ECM to the cell body on filopodia, which takes about 15 min. In a time window of 30 min, PsV-CD151 association occurs. (iii) In the samples recorded 30 – 180 min after CytD wash off, we observe that PsVs lose their HS-coat, and individual PsV-CD151 assemblies seem to merge into larger structures that are subsequently endocytosed. This model highlights the role of ECM interactions, actin dynamics, and receptor engagement in HPV16 entry.

Materials and Methods

Antibodies and PsVs

We used the following primary antibodies in immunostainings of proteins. For the capsid protein L1, a rabbit polyclonal antibody (pAb) K75 (diluted 1:1000) as described previously (Knappe et al., 2007 [\[32\]](#); Rommel et al., 2005 [\[33\]](#)), for CD151 a mouse monoclonal antibody (mAb) (1:200) (Biorad, cat# MCA1856GA), and for Itga6 a rabbit polyclonal antibody (1:200) (Invitrogen, cat# PA5-12334) were used. For Heparan sulfate (HS) a mouse IgM monoclonal antibody (1:200) (amsbio, cat# 370255-S) was used that reacts with an epitope in native heparan sulfate chains and not with hyaluronate, chondroitin or DNA, and poorly with heparin (mAb 10E4 (David et al., 1992 [\[34\]](#))). For HS neo-epitope (Yokoyama et al., 1999 [\[35\]](#)) detection, a mouse monoclonal antibody (1:200) (amsbio, cat#370260-S) was used that reacts only with heparitinase-treated heparan sulfate chains, proteoglycans, or tissue sections, and not with heparinase treated HSPGs. The antibody recognizes desaturated uronic acid residues (mAb 3G10 (David et al., 1992 [\[34\]](#))). As secondary antibodies we used a STAR GREEN-coupled goat-anti-rabbit (1:200) (Abberior, cat# STGREEN-1002-500UG), an

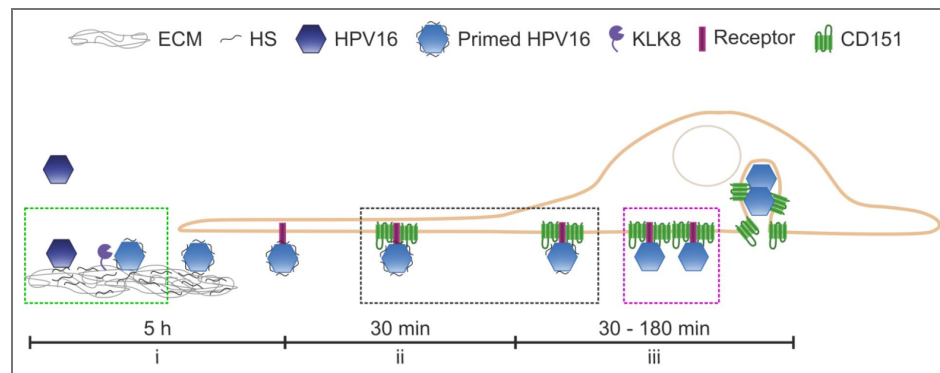


Figure 9. Model of ECM events, actin transport, and receptor engagement in HPV16 entry.

(i) During 5 h of incubation with CytD, the PsVs bind to HS of the ECM, are primed, and become coated with HS cleavage products, enabling them for cell surface receptor engagement. (ii) After CytD removal, within 15 min, HS-decorated viruses move along filopodia to the cell body and associate with CD151 assemblies (completed within 30 min), likely located at the filopodia. (iii) Eventually, they lose their HS coat, and individual HPV16-CD151 assemblies agglomerate into larger structures (platforms), which are subsequently endocytosed. Dashed rectangles mark PsVs representing populations as defined in Figure 7. Dashed green rectangle, PsVs with a distance to HS < 250 nm and to Itga6 > 250 nm. Dashed black rectangle, PsVs with a distance to HS < 250 nm and to Itga6 < 250 nm. Dashed magenta rectangle, PsVs with a distance to HS > 250 nm and to Itga6 < 250 nm.

AlexaFluor 594-coupled donkey-anti-rabbit (1:200) (Invitrogen, cat# A21207), an AlexaFluor 594-coupled donkey-anti-mouse (1:200) (Invitrogen, cat# A21203), an AlexaFluor 594-coupled donkey-anti-mouse IgM (1:200) (Invitrogen, cat# A21044), a STAR RED-coupled goat-anti-mouse (1:200) (Abberior, cat# STRED-1001-500UG) and a STAR RED-coupled goat-anti-rabbit (1:200) (Abberior, cat# STRED-1002-500UG). Additionally, phalloidin iFluor488 (1:1000 of ready to use solution) (abcam, cat# ab176753) or phalloidin iFluor647 (1:1000) (abcam, cat# ab176759) was used to stain F-actin. Staining of the PsV plasmid DNA was performed by click-labeling according to the manufacturer's instructions with the dye 6-FAM Azide (Baseclick EdU 488 kit, Carl Roth, cat# 1Y67.1).

HPV16 PsVs were produced following established procedures (Buck et al., 2004 [\[1\]](#); Spoden et al., 2012 [\[2\]](#)). HEK293TT cells were cultured in 175 cm² flasks and transfected with polyethylenimine (PEI), using equimolar amounts of codon-optimized HPV16 L1/L2 (pShell-16L1wt-16L2wt) and a reporter plasmid (pGL4.20-puro-HPV16 LCR (Wüstenhagen et al., 2018 [\[3\]](#))). For the production of 5-ethynyl-2'-deoxyuridine (EdU)-labeled PsVs, the culture medium was replaced 5 hours post-transfection with fresh medium containing 30 µM EdU. After 48 hours, cells were harvested, centrifuged, and washed twice with phosphate-buffered saline (PBS) supplemented with 9.5 mM MgCl₂ (1x PBS/ MgCl₂). After the final centrifugation, the pellet was resuspended in 1x PBS/ MgCl₂ containing 0.5% Brij58 (Sigma-Aldrich), and 250 units Benzonase (Merck Millipore), and incubated at 37°C for 24 hours on a rotating platform. Subsequently, lysates were chilled on ice and the NaCl concentration was adjusted to 0.85 M. After clarification by centrifugation, the supernatant was loaded onto an iodixanol (Optiprep) gradient consisting of 39%, 33%, and 27% layers (bottom to top). Gradients were equilibrated for 90 minutes at room temperature before ultracentrifugation at 55,000 rpm for 3.5 hours at 16°C. Fifteen fractions of 300 µl each were collected from the top and analyzed via luciferase reporter assay to identify peak fractions. PsV titers were determined based on packaged genomes (viral genome equivalents, vge) as previously described (Spoden et al., 2012 [\[2\]](#)). The concentrations of stock solutions were 7.7 x 10⁶ vge/µl (for microscopy experiments) and 14.1 x 10⁶ vge/µl (for the luciferase assay).

Cell culture

For microscopy, human immortalized keratinocytes (HaCaTs) (Cell Lines Services, Eppelheim, Germany) were cultured in high glucose Dulbecco's modified Eagle's medium (DMEM + GlutaMAX, Gibco, cat# 61965-026) supplemented with 10% FBS (PAN Biotech, cat# P30-3031) and 1% Penicillin/Streptomycin (10,000 U/ml Penicillin, 10 mg/ml Streptomycin; PAN Biotech, cat# P06-07100) at 37 °C with 5% CO₂. For the luciferase infection assay, cells were grown at 37°C and 5% CO₂ in Dulbecco's modified Eagle's medium (DMEM + GlutaMAX, Thermo Fisher Scientific), supplemented with 10% FBS (Sigma-Aldrich), 1% minimum essential medium non-essential amino acids (MEM non-essential amino acids (Thermo Fisher Scientific)), and 5 µg/ml ciprofloxacin (Fresenius Kabi). For experiments, the antibiotics in the medium were omitted.

Sample preparation and immunostaining

About 150,000 HaCaT cells were plated onto 25 mm diameter poly-L-lysine (PLL) coated (100 µg/ml PLL for 30 min) glass coverslips in 6-well plates and incubated for 24 h at 37°C and 5% CO₂. The next day, cells were incubated at 37°C and 5% CO₂ for 5 h with PsVs (50 vge/plated cell) and 10 µg/ml cytochalasin D (CytD; stock solution 10 mg/ml in dimethyl sulfoxide (DMSO)) (Life Technologies, cat# PHZ1063) or 30 µM (-)-blebbistatin (stock solution 13.68 mM in DMSO) (Sigma-Aldrich, cat# B0560-1MG) in DMEM supplemented with 10% FBS. For controls, we added the same amount of DMSO without CytD or blebbistatin. Cells were washed with PBS (137 mM NaCl, 2.7 mM KCl, 1.76 mM KH₂PO₄, 10 mM Na₂HPO₄, pH 7.4) and fresh DMEM supplemented with 10% FBS was added. Cells were incubated further at 37°C and 5% CO₂ for 0 min (here the medium was added followed by immediate removal), 15 min, 30 min, 60 min and 180 min.

In another set of experiments, about 150,000 HaCaT cells were plated onto 25 mm diameter PLL coated glass coverslips in 6-well plates and incubated for 24 h at 37°C and 5% CO₂. The next day, cells were incubated at 37°C and 5% CO₂ for 5 h with PsVs (50 vge/plated cell) and 10 µg/ml CytD

either together with 100 μ M leupeptin (stock solution 100 mM in ddH₂O) (Carl Roth, cat# CN33.1) or 5 μ M Furin inhibitor I (stock solution 5 mM in DMSO) (Sigma-Aldrich, cat# 344930-1MG) in DMEM supplemented with 10% FBS. For controls, we used only 10 μ g/ml CytD. Cells were washed with PBS and fresh DMEM supplemented with 10% FBS was added. Cells were incubated further at 37°C and 5% CO₂ for 0 min, 30 min and 60 min.

To analyze PsV binding to detached HaCaT cells, about 350,000 HaCaT cells per well were plated in 6-well plates and incubated for 24 h at 37°C and 5% CO₂. The next day, cells were detached by a 15 min incubation with 10 mM EDTA (in PBS, pH 7.4) at 37°C and 5% CO₂. Detached cells were collected and incubated with 50 vg/cell in DMEM supplemented with 10% FBS under constant rotation at 4 °C for 1 h. Cells were washed three times with PBS at 4 °C to remove any unbound PsVs. Then, cells were seeded onto 25 mm diameter PLL coated glass coverslips in 6-well plates and incubated for 1 h at 37°C and 5% CO₂.

Before staining, cells were washed twice with PBS and fixed at room temperature (RT) with 4% PFA in PBS for 30 min, unless membrane sheets were generated. In this case, the coverslips were placed in ice cold sonication buffer (120 mM KGlu, 20 mM KAc, 10 mM EGTA, 20 mM HEPES, pH 7.2) and a 100 ms ultrasound pulse at 100% power was applied. This was repeated until in total about 10 pulses were applied at different locations of the coverslip. Then, membrane sheets were fixed like cells at RT with 4% PFA in PBS for 30 min. PFA was removed, and residual PFA was quenched by 50 mM NH₄Cl in PBS for 30 min. Afterwards, samples were blocked with 3% BSA in PBS for 30 min. Staining of PsVs was performed by click-labeling with the dye 6-FAM for 30 min at RT according to the manufacturer's instructions. Then, samples were washed three times with PBS. In case of no PsV labeling by click-chemistry, directly after blocking, the respective primary antibodies were added: mouse IgM mAb against HS (1:200), mouse mAb against Δ -HS (1:200), rabbit pAb against Itga6 (1:200), mouse mAb against CD151 (1:200) and rabbit pAb K75 against L1 (1:1000) in 3% BSA in PBS for 2 h. Samples were washed three times with PBS before adding the respective secondary antibodies and fluorescent labelled phalloidins in 3% BSA in PBS for 1 h: for HS we added AlexaFluor 594 coupled to donkey-anti-mouse IgM (1:200), for Δ -HS AlexaFluor 594 coupled to donkey-anti-mouse (1:200), for Itga6 STAR RED coupled to goat-anti-rabbit (1:200), for CD151 AlexaFluor 594 coupled to donkey-anti-mouse (1:200) or STAR RED coupled to goat-anti-mouse (1:200), for L1 AlexaFluor 594 coupled to donkey-anti-rabbit (1:200) or STAR GREEN coupled to goat-anti-rabbit (1:200) and for F-actin phalloidin iFluor488 (1:1000) or phalloidin iFluor647 (1:1000). Afterwards, samples were washed three times with PBS, followed by mounting of the coverslips onto microscopy slides using ProLong Gold antifade mounting medium (Invitrogen, cat# P36930).

Confocal and STED microscopy

For confocal and STED microscopy, samples were imaged employing a 4-channel STED microscope from Abberior Instruments. The microscope is based on an Olympus IX83 confocal microscope equipped with an UPlanSApo 100x (1.4 NA) objective (Olympus, Tokyo, Japan). Confocal and STED micrographs were recorded simultaneously. For confocal imaging, a 488 nm laser was used for the excitation of 6-FAM, STAR GREEN and phalloidin iFluor488, and emission was detected at 500–550 nm. For STED imaging, a 561 nm laser (at 45%) was used for the excitation of AlexaFluor 594 (detection at 580–630 nm) and a 640 nm laser (at 45%) for the excitation of STAR RED and iFluor647-labelled phalloidin (detection at 650–720 nm), in combination with a 775 nm laser for depletion (at 45%). The pixel size was set to 25 nm, for overview images a pixel size of 250 nm was used.

Analysis of confocal and STED micrographs

Images were analyzed using Fiji ImageJ (Schindelin et al., 2012 [DOI](#)) in combination with a custom written macro, essentially as described previously (Schmidt et al., 2024 [DOI](#)). When analyzing double stainings of Alexa594 (red channel) and STAR RED (long red channel), we corrected for crosstalk from the red into the long red channel by subtracting 40% of the intensity of the red channel image from the long red channel image.

First, images were smoothed with a Gaussian blur to improve maxima detection. For confocal PsV-DNA micrographs, we employed a Gaussian blur of $\sigma = 1$, and for confocal PsV-L1 and STED micrographs a Gaussian blur of $\sigma = 0.5$. Local maxima were detected using the 'Find Maxima' function (noise tolerance 60 for L1 if stained together with phalloidin iFluor488, and 80 if stained together with CD151, HS, or phalloidin iFluor647, 3 for click-labeled PsVs, 8 for CD151, 6 for HS and 15 for Itga6), yielding maxima positions in pixel positions. For analysis, regions of interest (ROIs) were defined (for details see figure legends).

In order to define the cell border region (Supplementary Figure 3B), the ImageJ 'Make Binary' function was used on CD151 STED micrographs for generating a binary mask. When possible, the 'Wand' tool was used to further outline the area covered by the cell, otherwise the same was done manually using the original micrograph as a reference. From the outline in the binary mask, a ROI was created, that was then filled in white with the 'Flood Fill' Tool. The ROI was shrunk to a 20-pixel broad strip, and cleared outside of the strip, what leaves a 20-pixel broad, closed ribbon marking the intracellular side of the cell border that, however, exhibits arbitrary edges produced in the process above. These edges were removed by manual adjustments with the 'Pencil' tool and the clear function. Afterwards, the ROI defines the intracellular side of the cell border. The ROI was symmetrically broadened by 40 pixels (using the 'Enlarge' function). From the now at least 60 pixels broad ROI, approximately two thirds covered the intracellular and one third the extracellular side. In this ROI, using the ImageJ 'Find Maxima' function, PsV-L1 maxima were detected (at a noise tolerance of 80) with a custom written macro in ImageJ both in the cell border region ROI and in a ROI covering the entire micrograph (excluding a 2-pixel edge). From these values, the percentage of PsVs within the cell border region was calculated.

For measuring intensity over time, we measured in the ROI the mean gray value. For background correction, we subtracted the mean gray value measured in a ROI next to the cell.

For measuring maxima intensity, a ROI with a diameter of 125 nm (5 pixels) was placed onto the determined maxima positions (see above). Using these ROIs, the average mean gray value of each maximum was measured. The average background mean gray value was measured in a ROI placed next to the cell and subtracted from the average mean gray value of the maxima. For each membrane sheet, the average maxima intensity was calculated, followed by averaging of the values of the membrane sheets.

To measure the shortest distance, e.g., between PsV and CD151 maxima, as a quality control for the PsV maxima (CD151 maxima do not undergo the quality control step) we placed onto each maximum position a horizontal and a vertical linescan (31 pixels long $\times$ 3 pixels width). Afterwards, we fitted a Gaussian distribution to the intensity distribution of each maximum. Only if at least one of the fits exhibits a fit quality of $R^2 > 0.8$ and if the Gaussian maximum located central to the intensity distribution the maximum was included in the further analysis. Moreover, using a 125 nm ROI at the PsV and CD151 maxima positions, the center of mass of fluorescence was determined, yielding the maxima positions in sub-pixels. Based on these positions we further calculated the shortest distance of a PsV maximum to the nearest CD151 maximum.

Using the shortest distances, we either calculated the fraction of closely associated PsVs (PsVs with a distance ≤ 80 nm to e.g., CD151) over time, or the average shortest distance over time. The fraction of closely associated PsVs was corrected for random background association that increases with the maxima density. The relationship between background association and maxima density was expressed by a linear equation obtained from the same images re-analyzed after flipping them horizontally and vertically (see e.g., Supplementary Figure 7).

The Pearson correlation coefficient (PCC) was calculated with a custom written macro in ImageJ. As a control, we performed the analysis on horizontally and vertically flipped ROI-defined images as well (see e.g., Supplementary Figure 6A). The cell body PCC between PsVs and HS is measured in smaller ROIs covering exclusively the cell body region (this is confirmed with the Itga6 or F-actin image as reference).

Per condition and biological replicate, 14 - 15 images were analyzed and values were averaged. For images from one set of experiments, in the figure the same channels are shown at the same settings of brightness and contrast, if not stated otherwise.

Epi-fluorescence microscopy and image analysis

For epifluorescence microscopy, we used microscopic equipment and settings as previously described (Homsí et al., 2014 [\[1\]](#)) except of the illumination system, which was replaced by a SPECTRA X Light engine; Lumencor, Beaverton, OR, USA). In brief, PFA-fixed cells were imaged in a chamber filled with 1 ml of PBS to which 50 μ l of a saturated solution of TMA-DPH [1-(4-trimethyl-ammonium-phenyl)-6-phenyl-1,3,5-hexatriene-p-toluenesulfonate (T204; Thermofisher)] in PBS was added, for visualizing the membranes in the blue channel. PsVs, HS and Itga6 were imaged in the green, red and far-red channels, respectively.

For analysis, rectangular ROIs of $\approx 45 \times 10^3$ pixel were placed onto the cell body such that approximately one half of the ROI covers the ECM area and the other one the cell body. The cell periphery was defined using the TMA-DPH image from which a binary mask was created. With reference to the binary mask, a band of up to 30 pixels width was created beginning at the cell body and reaching out towards the cell periphery. Using this ROI, in the PsV-DNA image the size of the ROI and the mean gray values were measured from which the integrated intensities were calculated after background correction.

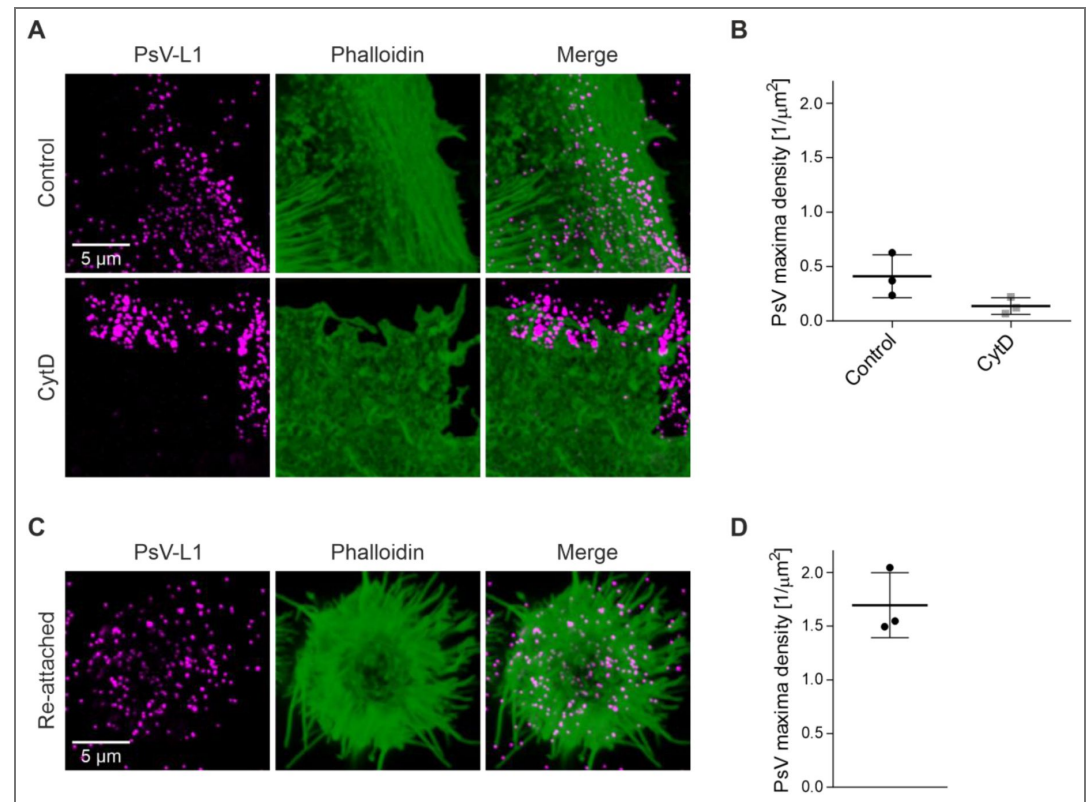
Luciferase infection assay

HaCaT cells were grown in 24-well plates and allowed to adhere overnight. The following day, cells were treated with HPV16 PsVs at a concentration of approximately 100 vge per cell, in the presence or absence of 10 μ g/ml cytochalasin D (CytD) in DMSO or an equivalent amount of DMSO (Control). In one condition, PsVs/CytD were applied for 5 hours, after which the medium was replaced with fresh medium lacking the compounds, and incubation was continued for additional 19 hours (in total 24 h). In another condition, cells were exposed to PsVs/CytD continuously for the full 24 h period. Then, cells were washed with PBS and lysed using 1 x Cell Culture Lysis Reagent (Promega, cat# E153A). Following high-speed centrifugation, luciferase activity in the cleared lysates was quantified using an LB 942 Tristar 3 Multimode Microplate Reader (Berthold Technologies). Cytotoxic effects, accompanied by a loss of membrane integrity (indicated by released lactate dehydrogenase (LDH)), were determined by measuring the LDH levels in the cell lysates. LDH activity was assessed using the CytoTox-ONE Homogeneous Membrane Integrity Assay (Promega, cat# G7891). LDH fluorescence was quantified according to the manufacturer's instructions using the LB 942 Tristar 3 Multimode Microplate Reader.

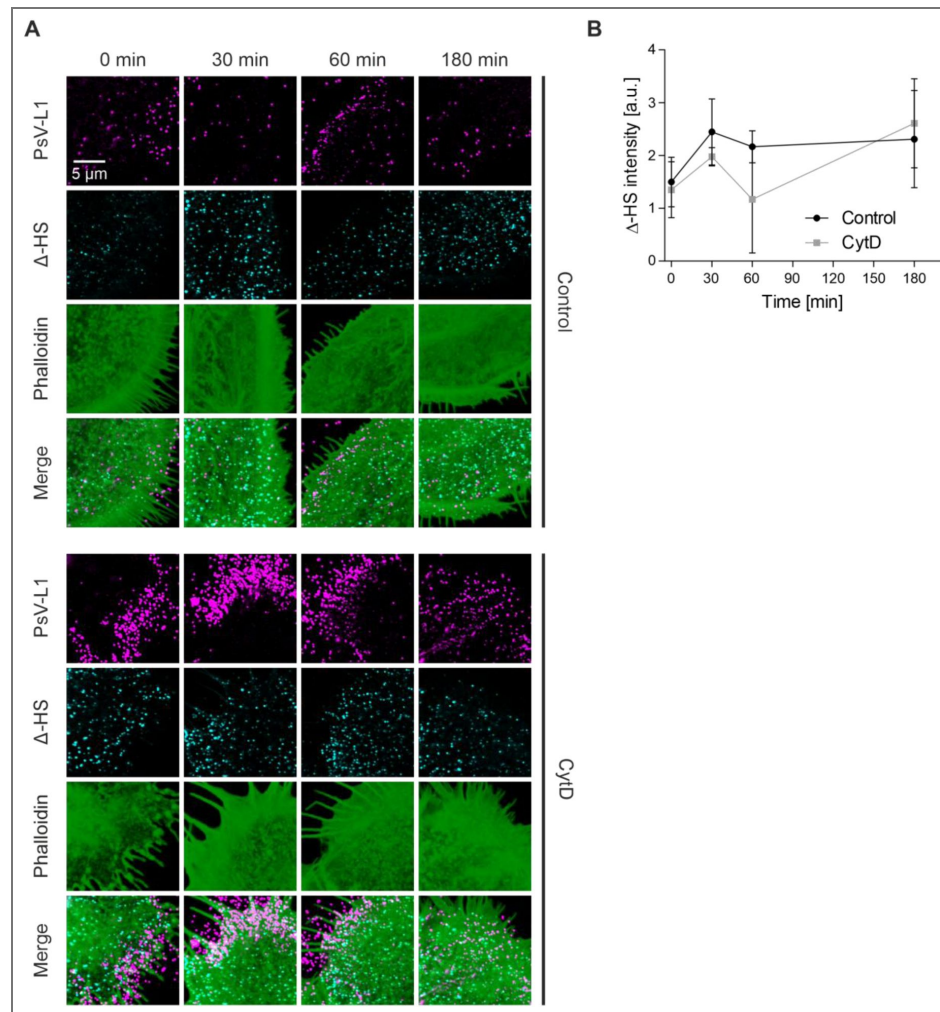
Statistics

Microscopy data sets were based on three biological replicates. One replicate includes per time point the average of 14 - 15 images analyzed (with exception of Figure 4 [\[1\]](#), which includes 14 - 35 analyzed cells per replicate and time point). For the infection assay, data sets were based on three biological replicates. One biological replicate is the average of three technical replicates. Data was tested for significance with a two-tailed, unpaired student's *t*-test with significance * = $p < 0.05$, ** = $p < 0.01$ and *** = $p < 0.001$.

Supplementary information

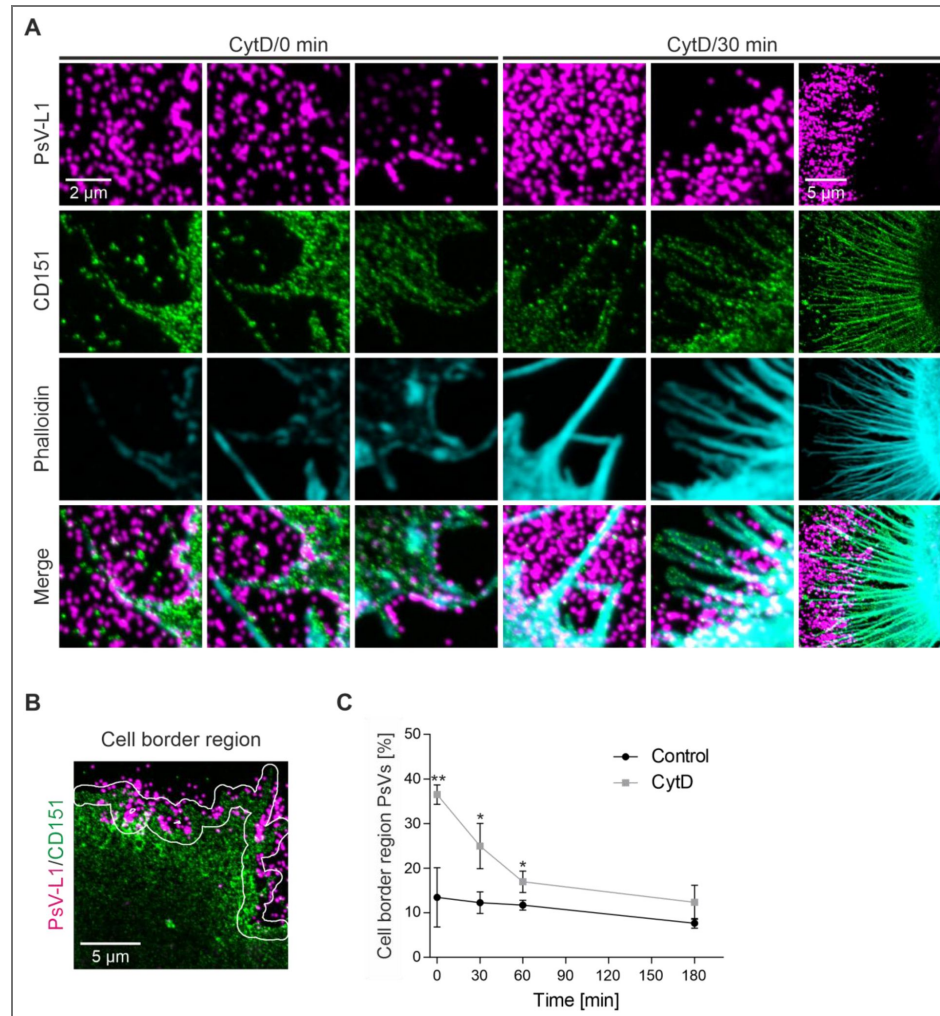


Supplementary Figure 1. PsV density at the adhered membrane of substrate-adhered grown and re-attached HaCaT cells. (A) The shown HaCaT cells are from the 0 min time point (top, Control; bottom, CytD) of the experiment described in Figure 5. Only the PsV-L1 (STAR GREEN; magenta LUT) and the F-actin phalloidin (iFluor647; green LUT) staining are shown. (B) In order to count the PsVs at the basal membrane, we placed ROIs covering only the cell body. Within these ROIs the PsV maxima were counted and related to the size of the analyzed area. Values are given as means $\pm$ SD ($n = 3$ biological replicates). Statistical differences between Control and CytD were analyzed by using the two-tailed, unpaired student's t -test ($n = 3$), but the analysis yielded no p -value below 0.05. (C) HaCaT cells were detached by a 15 min incubation with 10 mM EDTA (in PBS) and incubated with PsVs at 4 °C for 1 h under constant rotation. After washing the cells three times with PBS, they were seeded onto PLL-coated glass coverslips for 1 h. Then, cells were washed, fixed and stained for L1 (STAR GREEN; magenta LUT) by indirect immunofluorescence and for F-actin by iFluor647-labelled phalloidin (green LUT). PsV-L1 and F-actin staining were imaged as in (A) in the confocal mode of a STED microscope. (D) Analysis as in (B). Values are given as means $\pm$ SD ($n = 3$ biological replicates).



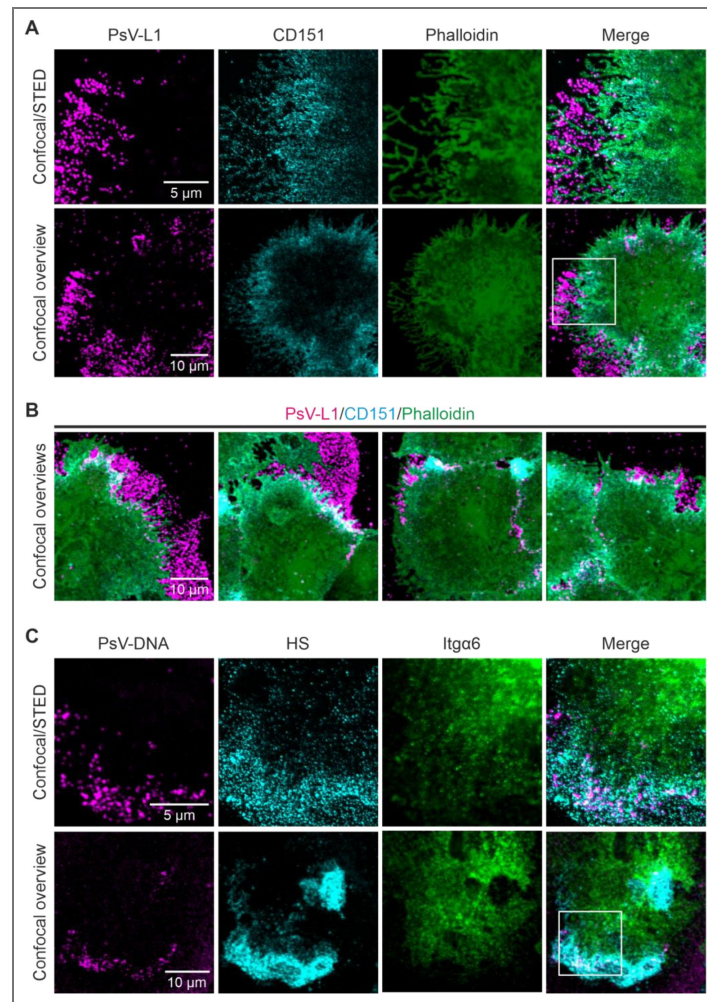
Supplementary Figure 2. HS neo-epitope (Δ-HS) staining intensity after CytD treatment.

(A) HaCaT cells were incubated with PsVs at 37 °C for 5 h, in the absence (Control, upper panels) or presence of 10 µg/ml CytD (CytD, lower panels). Afterwards, cells were washed and incubated without PsVs/CytD further for up to 180 min, before they were fixed and stained. Immunofluorescence was used for the visualization of L1 (STAR GREEN; magenta LUT) and the HS neo-epitope, referred to as Δ-HS (AlexaFluor 594; cyan LUT). F-actin was stained by iFluor647-labelled phalloidin (green LUT). PsV-L1, Δ-HS and F-actin staining were imaged in the confocal mode of a STED microscope. For analysis, we placed large ROIs onto the images that covered mainly the cell body but included parts of the cell periphery as well. (B) The mean Δ-HS intensity over time. Values are given as means ± SD (n = 3 biological replicates). Statistical differences between the same time points of Control and CytD were analyzed by using the two-tailed, unpaired student's *t*-test (n = 3), but the analysis yielded no p-values below 0.05.



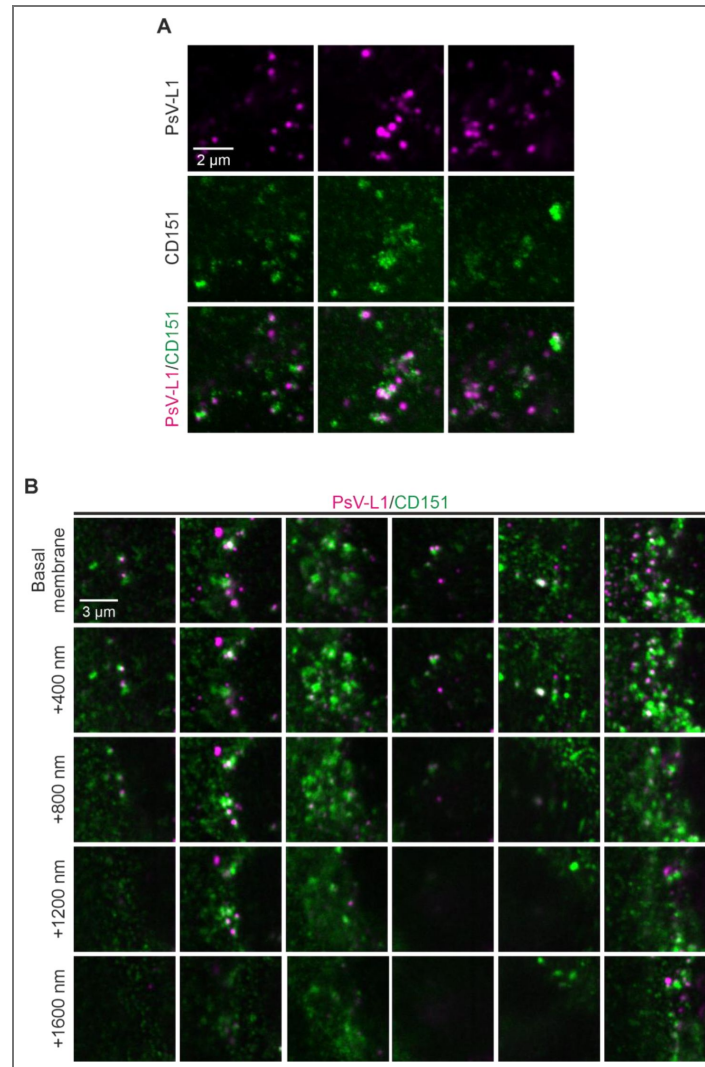
Supplementary Figure 3. Variability of filopodia/diminishment of PsVs from the cell border region after CytD removal.

(A) More images from the experiment described in Figure 5. We observed that many cells exhibit filopodia. Due to the large variability in number and shape we cannot show a representative image. Instead, we present for the CytD condition (0 min and 30 min) a variety of cells with CD151 positive filopodia. (B) Based on the CD151 image (green LUT), a cell border strip is defined and broadened by 20 pixels on each side, in the following referred to as the cell border region (for details see materials and methods). Please note that the cell border region is different from the cell periphery described in Figure 4 that delineates the cell body from the periphery. Instead, the cell border region covers both sides of the cell border (approximately two thirds inside and one third outside of the cell; for details see methods). (C) Diminishment of PsV maxima from the cell border region over time, expressed as percentage of all PsVs present in the image. Values are given as means $\pm$ SD ($n = 3$ biological replicates). Statistical differences between the same time points of Control and CytD were analyzed by using the two-tailed, unpaired student's *t*-test ($n = 3$, for details see materials and methods).



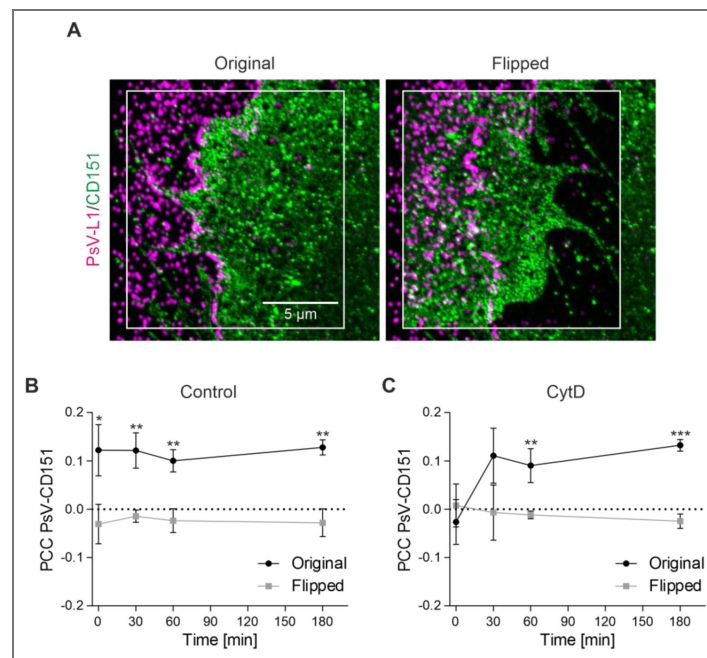
Supplementary Figure 4. Overview images.

(A) Upper row, micrographs from Figure 5 [\[5\]](#), CytD/0min, showing as well F-actin staining (green LUT) and CD151 using the cyan LUT. Lower row, confocal overview. After taking the overview, the area in the white box was imaged as described in the legend of Figure 5 [\[5\]](#), yielding the images shown in the upper row in A. (B) More overview images from the experiment described in Figure 5 [\[5\]](#), condition CytD/0 min, showing examples of large and small PsV accumulations. (C) Upper row, micrographs from Figure 6 [\[6\]](#), CytD/0 min. Lower row, confocal overview. The area in the white box was imaged as described in the legend of Figure 6 [\[6\]](#), yielding the images of the upper row in C.



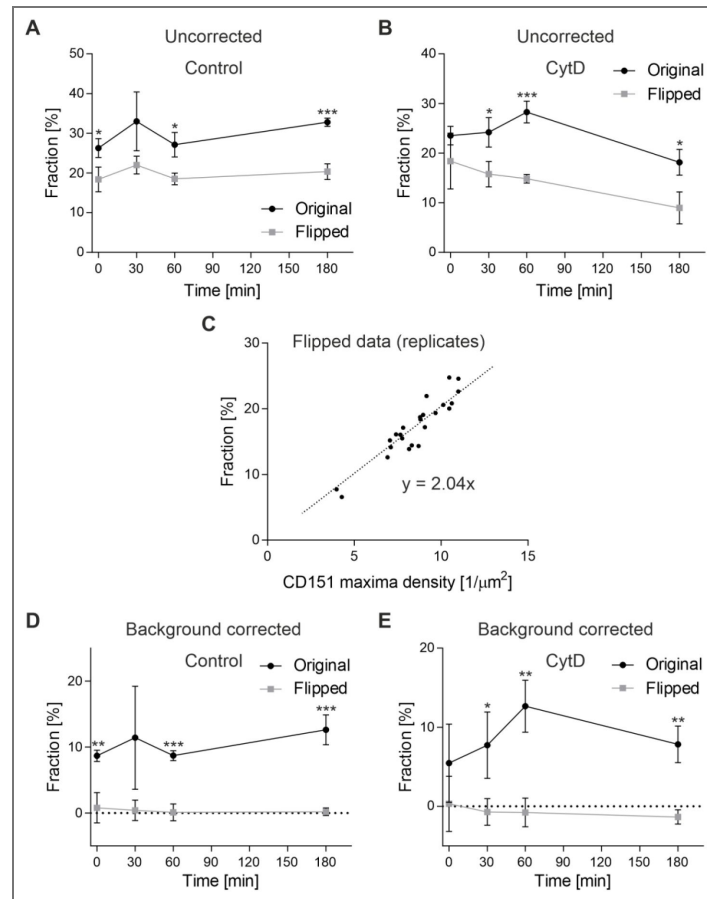
Supplementary Figure 5. Examples of agglomerated CD151 maxima associated with PsVs that presumably represent endocytic structures.

(A) From the CytD/180 min time point described in Figure 5 [\[1\]](#), we show more examples of agglomerated CD151 maxima (green LUT, see green patches) that associate with PsVs (magenta LUT). (B) Same experiment as in (A). Starting at the basal membrane, at confocal resolution we imaged CD151 (green LUT) and PsV-L1 (magenta LUT). Cells were further scanned by 400 nm steps in the axial direction. Some of the CD151 agglomerates noticed in the basal membranes appear to continue deeper into the cell, more than a micrometer in the second example from the left. We propose that the agglomerated CD151 maxima close to PsVs feature the characteristics of endocytic structures, as CD151 has been shown to co-internalize with PsVs (Scheffer et al., 2013 [\[2\]](#)), and as these structures invaginate into the cell, like PsV filled tubular organelles previously described by electron microscopy (Schelhaas et al., 2012 [\[3\]](#)). Images in (A) and (B) are shown at the same settings of brightness and contrast but at different settings compared to Figure 5 [\[1\]](#).



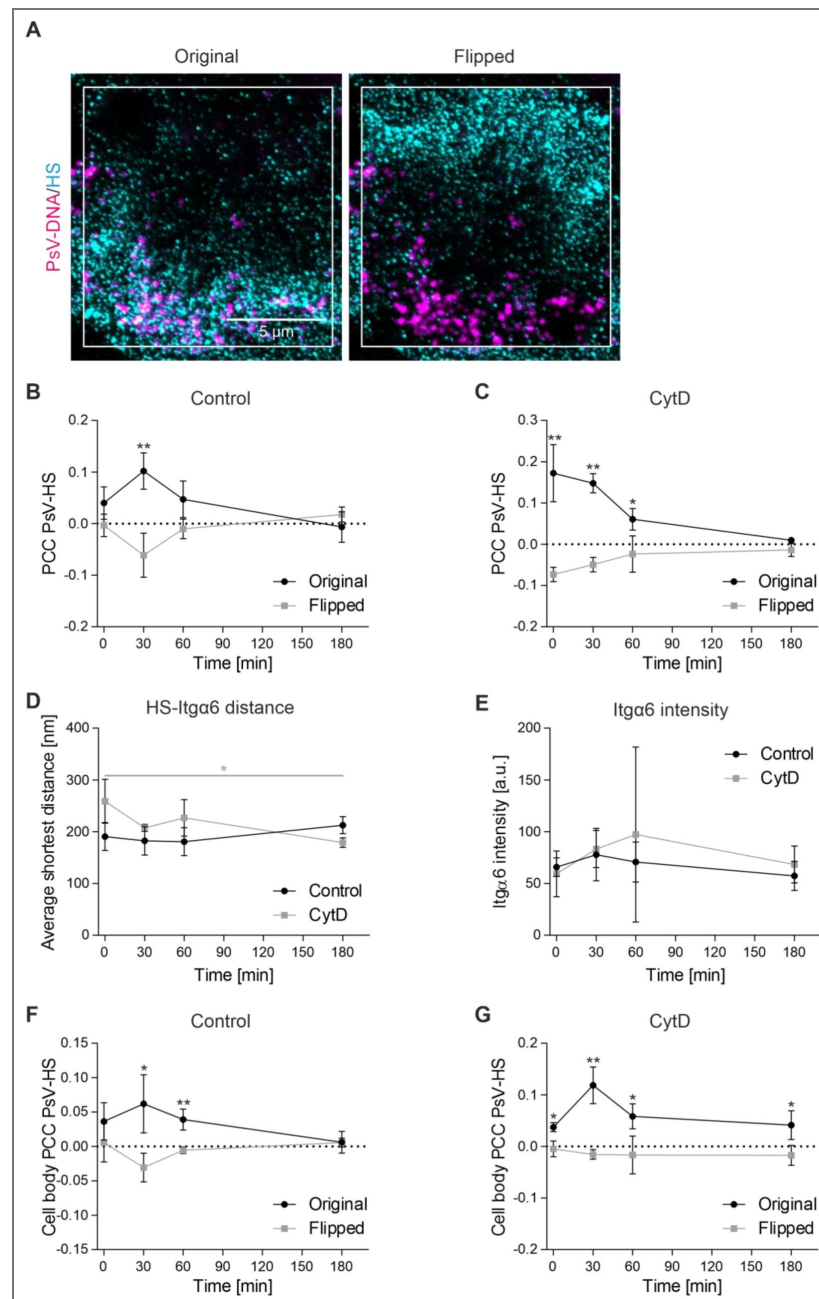
Supplementary Figure 6. PCC values of flipped images.

(A) For each image pair, we determined the PCC on respective non-overlapping images that have the same density of objects and intensities. To this end, one channel was flipped horizontally and vertically. Left, original images (from the CytD/0 min condition). The ROI for analysis is illustrated as white box. Right, in the ROI, the CD151 image (green LUT) was flipped horizontally and vertically. (B + C) Shown are the PCCs over time of Figure 5C obtained on the original images ((B) Control, (C) CytD) together with the PCCs of the flipped images. Values are given as means $\pm$ SD ($n = 3$ biological replicates). Statistical differences between the same time points of original and flipped images was analyzed by using the two-tailed, unpaired student's t -test ($n = 3$, for details see methods).



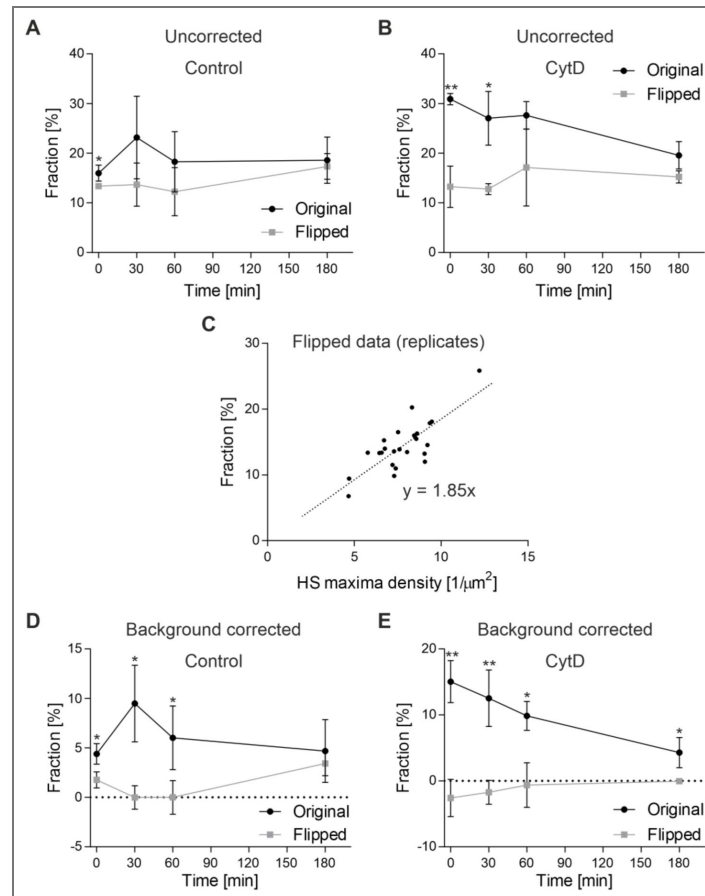
Supplementary Figure 7. Background correction of the fraction of PsVs closely associated with CD151.

(A) The fraction of closely associated PsVs (PsV-L1 maxima with a distance ≤ 80 nm to the next nearest CD151 maximum) in the Control of Figure 5 was analyzed on original and flipped images (for an example of a flipped image see Supplementary Figure 6A). On flipped images, we often find values more than half of the values of the original images, demonstrating that many PsVs have a distance ≤ 80 nm to CD151 merely by chance (background association). (B) Same as (A), for CytD. (C) Each time point in (A) and (B) obtained from flipped images is the average of three biological replicates. We use these altogether 24 data points, plotting the fraction of closely associated PsVs against the CD151 maxima density. The fraction increases with the maxima density, as the chance of random association increases with the maxima density. The fitted linear regression line describes the dependence of the background association from the maxima density. As a result, the background association (y) can be calculated for any maxima density (x) in original images with the equation $y = 2.04x$. Please note that the CytD/0 min may be overcorrected as we subtract background association with reference to the CD151 maxima density of the entire ROI (for an example ROI see Supplementary Figure 6A), although the local maxima density at distal PsVs is lower. On the other hand, PsVs at the cell border may have a larger local CD151 maxima density and consequently are undercorrected. (D) and (E), for each original and flipped image of (A) and (B), the background was calculated and subtracted from the uncorrected value. The original background corrected values in (D) and (E) are shown in Figure 5D. Values are given as means $\pm$ SD ($n = 3$ biological replicates). Statistical differences between the same time points of original and flipped images were analyzed by using the two-tailed, unpaired student's *t*-test ($n = 3$, for details see methods).



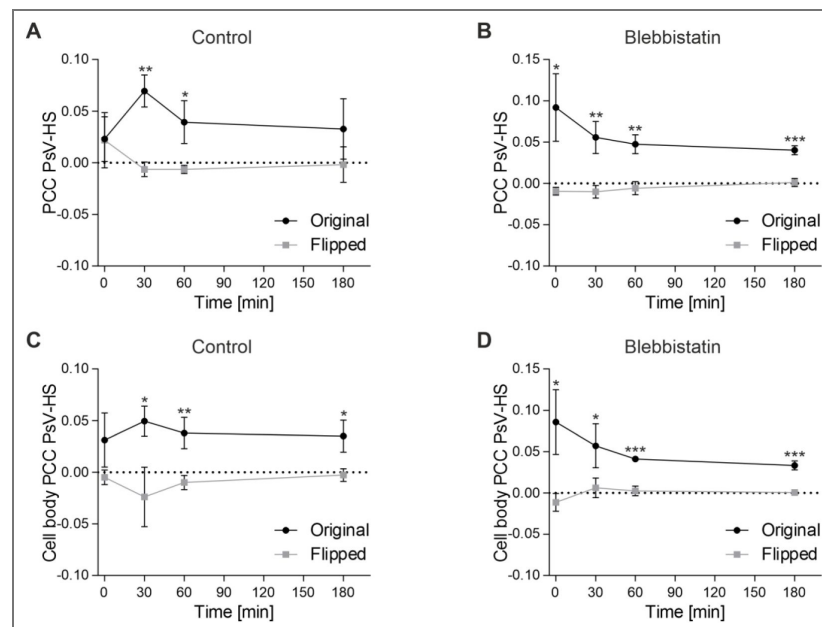
Supplementary Figure 8. PCC values of flipped images, HS-Itga6 shortest distance and Itga6 intensity.

(A) For each image pair, we determined the PCC on respective non-overlapping images that have the same density of objects and intensities. To this end, one channel was flipped horizontally and vertically. Left, original images (from the CytD/0 min condition). The ROI for analysis is illustrated as white box. Right, in the ROI, the HS image (cyan LUT) was flipped horizontally and vertically. (B + C) Shown are the PCCs over time of Figure 6C obtained with the original images ((B) Control, (C) CytD) together with the PCCs of the flipped images. (D) The shortest distance of HS maxima to the next nearest Itga6 maxima over time. (E) The mean Itga6 intensity over time. (F + G) Shown are the PCCs over time of Figure 6D obtained with the original images ((F) Control, (G) CytD) together with the PCCs of the flipped images. Values are given as means $\pm$ SD ($n = 3$ biological replicates). Statistical differences between the same time points of original and flipped images ((B), (C), (F) and (G)) or Control and CytD (E). (D) Difference between CytD/0 min and CytD/180 min. Statistical differences were analyzed by using the two-tailed, unpaired student's t -test ($n = 3$, for details see methods).



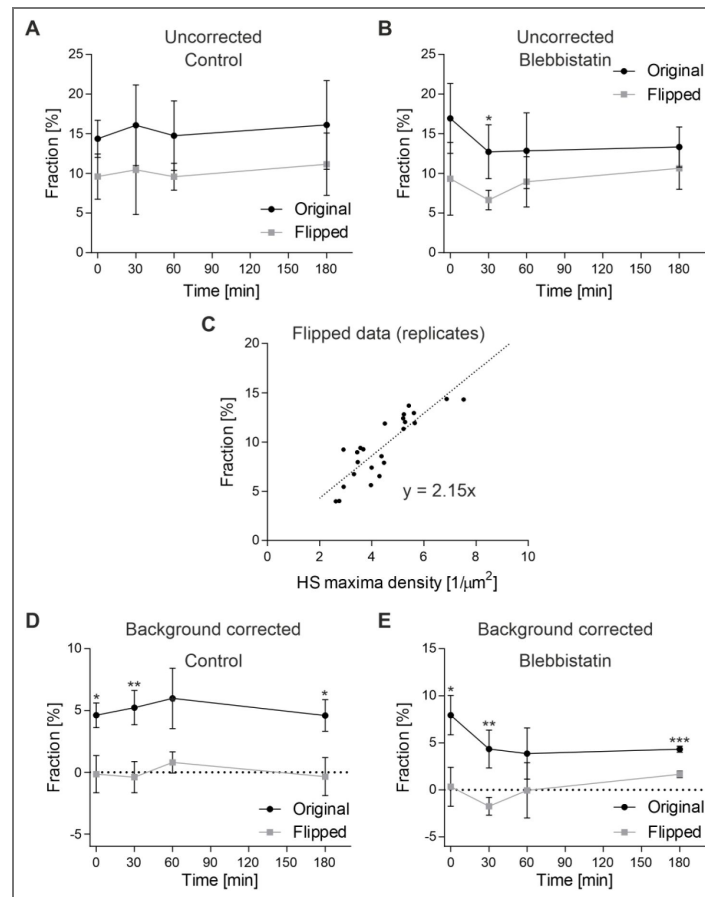
Supplementary Figure 9. Background correction of the fraction of PsVs closely associated with HS.

(A) The fraction of closely associated PsVs (PsV-DNA maxima with a distance ≤ 80 nm to the next nearest HS maximum) in the Control of Figure 6 was analyzed on original and flipped images (for an example of a flipped image see Supplementary Figure 8A). (B) Same as (A), for CytD. (C) From the altogether 24 analyzed flipped images, we obtained the equation $y = 1.85x$ for calculating the background association (please see Supplementary Figure 7 for more details). (D) and (E), background corrected graphs of (A) and (B). For each original and flipped image the background was calculated and subtracted from the uncorrected value. The original background corrected values in (D) and (E) are shown in Figure 6E. Values are given as means $\pm$ SD ($n = 3$ biological replicates). Statistical differences between the same time points of original and flipped images were analyzed by using the two-tailed, unpaired student's t -test ($n = 3$, for details see methods).



Supplementary Figure 10. PCC values of flipped images.

(A) and (B) The PCC values between PsV and HS of Figure 8C (A) Control, (B) Blebbistatin) plotted together with the respective PCC values of flipped images. (C) and (D) The cell body PCC values between PsV and HS of Figure 8D (C) Control, (D) Blebbistatin) plotted together with the respective PCC values of flipped images. Values are given as means $\pm$ SD ($n = 3$ biological replicates). Statistical differences between the same time points of original and flipped images were analyzed using the two-tailed, unpaired student's t -test ($n = 3$, for details see methods).



Supplementary Figure 11. Background correction of the fraction of PsVs closely associated with HS upon blebbistatin treatment.

(A) The fraction of closely associated PsVs (PsV-L1 maxima with a distance ≤ 80 nm to the next nearest HS maximum) in the Control of Figure 8 was analyzed on original and flipped images. (B) Same as (A), for blebbistatin. (C) From the altogether 24 analyzed flipped images, we obtained the equation $y = 2.15x$ for calculating the background association (please see Supplementary Figure 7 for more details). (D) and (E), background corrected graphs of (A) and (B). For each original and flipped image the background was calculated and subtracted from the uncorrected value. The original background corrected values in (D) and (E) are shown in Figure 8E. Values are given as means $\pm$ SD ($n = 3$ biological replicates). Statistical differences between the same time points of original and flipped images were analyzed by using the two-tailed, unpaired student's t -test ($n = 3$, for details see methods).

	0 min			30 min			60 min			180 min		
	Control [%]	CytD [%]	p-value	Control [%]	CytD [%]	p-value	Control [%]	CytD [%]	p-value	Control [%]	CytD [%]	p-value
HS < 250 nm & Itga6 < 250 nm	62.31 ± 5.05	47.83 ± 3.26	0.0271	65.48 ± 8.35	63.67 ± 1.14	0.7764	59.06 ± 10.64	67.16 ± 9.84	0.4739	63.69 ± 4.27	59.28 ± 3.17	0.3065
HS < 250 nm & Itga6 > 250 nm	12.18 ± 5.51	46.04 ± 3.67	0.0019	15.46 ± 3.14	26.12 ± 2.16	0.0167	13.25 ± 1.94	19.87 ± 3.12	0.0637	11.97 ± 1.53	10.64 ± 1.15	0.3808
HS > 250 nm & Itga6 < 250 nm	21.59 ± 4.07	4.34 ± 0.37	0.0040	15.83 ± 5.73	8.58 ± 2.33	0.1723	23.49 ± 6.55	9.60 ± 4.81	0.0731	20.88 ± 3.31	25.24 ± 1.96	0.1842
HS > 250 nm & Itga6 > 250 nm	3.93 ± 0.30	1.79 ± 0.60	0.0110	3.23 ± 0.88	1.63 ± 0.21	0.0675	4.19 ± 2.43	3.38 ± 2.20	0.7425	3.46 ± 0.54	4.84 ± 2.87	0.5393

Supplementary Table 1. Fraction of PsVs in percent of each of the four distance categories (see left column).

Values are means ± SD of the data shown in Figure 7B. For each time point and category, p-values between Control and CytD were calculated by using the two-tailed, unpaired student's *t*-test (*n* = 3 biological replicates). P-values < 0.05 are illustrated in bold.

Data availability

On reasonable request, data of the current study are available from the corresponding authors.

Acknowledgements

LF and TL were funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation), Projektnummer 322863883 (LF and TL) and Projektnummer 270976260 (TL).

Additional information

Funding

Funder	Grant reference number	Author
Deutsche Forschungsgemeinschaft (DFG)	Projektnummer 322863883	Luise Florin Thorsten Lang
Deutsche Forschungsgemeinschaft (DFG)	Projektnummer 270976260	Thorsten Lang

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Peer reviews

Reviewer #1 (Public review):

Summary:

The author's goal was to arrest PsV capsids on the extracellular matrix using cytochalasin D. The cohort was then released and interaction with the cell surface, specifically with CD151 was assessed.

The model that fragmented HS associated with released virions mediates the dominant mechanism of infectious entry has only been suggested by research from a single laboratory and has not been verified in the 10+ years since publication. The authors are basing this study on the assumption that this model is correct, and these data are referred to repeatedly as the accepted model despite much evidence to the contrary. The discussion in lines 65-71 concerning virion and HSPG affinity changes is greatly simplified. The structural changes in the capsid induced by HS interaction and the role of this priming for KLK8 and furin cleavage has been well researched. Multiple laboratories have independently documented this. If this study aims to verify the shedding model, additional data needs to be provided.

Note on revisions:

The authors did an excellent job in their revision to include data from the effect of proteolytic priming on their observed virion transfer to the cell body. All other minor issues were addressed adequately.

The work could be especially critical to understanding the process of in vivo infection. <https://doi.org/10.7554/eLife.107139.2.sa2>

Reviewer #2 (Public review):

The study design involves infecting HaCaT cells (immortalised keratinocytes mimicking basal cells of a target tissue) and observing virus localization with and without actin polymerization inhibition by cytochalasin D (cytoD) to analyze virion transfer from the ECM to the cell via filopodial structures, using cellular proteins as markers.

In the context of the model system, the authors stress in the revised version the importance of using HaCaT cells as a relevant 'polarized' cell model for infection. The term 'polarized' is used in the cell biological literature for epithelial cells to describe a strict apical vs. basolateral demarcation of the plasma membrane with an established diffusion barrier of the tight junction. However, HaCat cells do not form tight junctions. In squamous epithelia, such barriers are only found in granular layers of the epithelium. The published work cited in support of their claims either does not refer to polarity or only in the context of other cells such as CaCo-2 cells.

Overall, the matter of polarity would be important, if indeed the virus could only access cell-associated HSPGs as primary binding receptor, or the elusive secondary receptor via the ECM in the used model system (HaCaT cells), if they would locate exclusively basolaterally. This is

at least not the case for binding, as observed in several previous publications (just two examples: Becker et al, 2018, Smith et al., 2008). With only a rather weak attempt at experimental verification of their model system with regards to polarity of binding, the authors then go on to base their conclusions on this unverified assumption.

This is one example of several in the manuscript, where claims for foundational premises, observations, and/or conclusions remain undocumented or not supported by experimental data.

Another such example is the assumption of transfer of the virus from ECM to the tetraspanin CD151. Here, the conclusions are based on the poorly documented inability of the virus to bind to the cell body, which is in stark contrast to several previous publications, and raises questions. Thus, association with CD151 likely occurs both from ECM derived virus AND virus that binds to cells, so that any conclusions on the mode of association is possible only in live cell data (which is not provided). Overall, their proposed model thus remains largely unsubstantiated with regards to receptor switching.

There are a number of important additional issues with the manuscript:

First, none of the inhibitors have been tested in their system for efficacy and specificity, but rely on published work in other cell types. This considerably weakens the confidence on the conclusion drawn by the authors.

Second, the authors aim to study transfer from ECM to the cell body and effects thereof. However, there are still substantial amounts of viruses that bind to the cell body compared to ECM-bound viruses in close vicinity to the cells. This is in part obscured by the small subcellular regions of interest that are imaged by STED microscopy, or by the use of plasma membrane sheets. This remains an issue despite the added Supple. Fig. 1, where also only sub cellular regions are being displayed. As a consequence the obtained data from time point experiments is skewed, and remains for the most part unconvincing, largely because the origin of virions in time and space cannot be taken into account. This is particularly important when interpreting the association with HS, the tetraspanin CD151, and integral alpha 6, as the low degree of association could be originating from cell bound and ECM-transferred virions alike.

Third, the use of fixed images in a time course series also does not allow to understand the issue of a potential contribution of cell membrane retraction upon cytoD treatment due to destabilisation of cortical actin. Or, of cell spreading upon cytoD washout. The microscopic analysis uses an extension of a plasma membrane stain as marker for ECM bound virions, this may introduce a bias and skew the analysis.

Fourth, while the use of randomisation during image analysis is highly recommended to establish significance (flipping), it should be done using only ROIs that have a similar density of objects for which correlations are being established. For instance, if one flips an image with half of the image showing the cell body, and half of the image ECM, it is clear that association with cell membrane structures will only be significant in the original. But given the high density of objects on the plasma membrane, I am not convinced that doing the same by flipping only the plasma membrane will not also obtain similar numbers than the original.

<https://doi.org/10.7554/eLife.107139.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

The authors' goal was to arrest PsV capsids on the extracellular matrix using cytochalasin D. The cohort was then released, and interaction with the cell surface, specifically with CD151, was assessed.

The model that fragmented HS associated with released virions mediates the dominant mechanism of infectious entry has only been suggested by research from a single laboratory and has not been verified in the 10+ years since publication. The authors are basing this study on the assumption that this model is correct, and these data are referred to repeatedly as the accepted model despite much evidence to the contrary.

We stated in the introduction on line 65/66 Two release mechanisms are discussed, that mutually are not exclusive. This implies that we do not consider the shedding model as 'the accepted model'. Furthermore, we do not state in the discussion neither that the shedding model is the preferred one. However, we referred to the shedding model in the discussion, because we find HS associated with transferred PsVs, which is in line with this model.

The discussion in lines 65-71 concerning virion and HSPG affinity changes is greatly simplified. The structural changes in the capsid induced by HS interaction and the role of this priming for KLK8 and furin cleavage have been well researched. Multiple laboratories have independently documented this. If this study aims to verify the shedding model, additional data need to be provided.

Our findings are compatible with both models, and we do not aim to verify the shedding model neither want to disprove the priming model. However, as we understand, the referee wishes more visibility of the priming model. Therefore, using inhibitors previously used in the field, we tested whether inhibition of KLK8 or furin reduces PsV translocation to the cell body (after CytD wash off). Leupeptin blocks transport, while Furin inhibitor I still allows some initial translocation. We incorporated this new data as Figure 2 (line 265): "...we would expect that inhibition of L1 processing during the CytD incubation prevents the recovery of PsV translocation from the ECM to the cell body (Figure 2A and D). To test for this possibility, as employed in earlier studies, the protease inhibitor leupeptin was used to inhibit proteases including KLK8 which is required for L1 cleavage (Cerqueira et al. 2015). Employing this inhibitor, the PCC between PsV-L1 and F-actin staining remains negative after CytD removal, showing that for translocation indeed the action of proteases is required (Figure 2B and D). In contrast, inhibition of L2 cleavage by a furin specific inhibitor has no effect on the PCC (Figure 2C and D). However, it should be noted that we occasionally observe PsVs not completely translocating but accumulating at the border of the F-actin stained area (for example see Figure 2C (60 min)). This results in an increase of the PCC almost equal to complete translocation, explaining why the PCC remains unaffected despite a furin inhibitory effect. Hence, furin inhibition may have some effect on translocation that, however, is undetected in this type of analysis.'

Moreover, we have added a paragraph discussing how our data integrates into the established model of the HPV infection cascade (line 604): 'HPV infection is the result of several steps, starting with the initial binding of virions via electrostatic and polar interactions (Dasgupta et al. 2011) to the primary attachment site HS (Richards et al. 2013), which induces capsid modification (Feng et al. 2024; Cerqueira et al. 2015) and HS cleavage (Surviladze et al. 2015), enabling the virion to be released from the ECM or the glycocalyx. Next, virions bind to the cell surface to a secondary receptor complex that forms over time, and become internalized via endocytosis, before they are trafficked to the nucleus (Ozbun and Campos 2021; Mikulić et al. 2021). Regarding the transition from the primary attachment site to cell surface binding, as already outlined in the introduction, two models

are discussed. In one model, proteases cleave the capsid proteins. After priming, the capsids are structurally modified and the virion can dissociate from its HS attachment site. It has been suggested that capsid priming is mediated by KLK8 (Cerqueira et al. 2015) and furin (Richards et al. 2006). In our system, KLK8 inhibition blocks PsV transport, while furin inhibition has some effect that, however, cannot be detected in this analysis (Figure 2) suggesting furin engagement at later steps in the infection cascade. This is in line with earlier in vitro studies on the role of cell surface furin (Surviladze et al. 2015; Day et al. 2008; Day and Schiller 2009). In any case, our results align with both models of ECM detachment: one involving HS cleavage (HS co-transfer) and another involving capsid modification (by e.g., KLK8).’

The model should be fitted into established entry events,...

Please see our reply above.

or at minimum, these conflicting data, a subset of which is noted below, need to be acknowledged.

(1) The Sapp lab (Richards et al., 2013) found that HSPG-mediated conformational changes in L1 and L2 allowed the release of the virus from primary binding and allowing secondary receptor engagements in the absence of HS shedding.

(2) Becker et al. found that furin-precleaved capsids could infect cells independently of HSPG interaction, but this infection was still inhibited with cytochalasin D.

(3) Other work from the Schelhaas lab showed that cytochalasin D inhibition of infection resulted in the accumulation of capsids in deep invaginations from the cell surface, not on the ECM

(4) Selinka et al., 2007, showed that preventing HSPG-induced conformational changes in the capsid surface resulted in noninfectious uptake that was not prevented with cytochalasin D.

(5) The well-described capsid processing events by KLK8 and furin need to be mechanistically linked to the proposed model. Does inhibition of either of these cleavages prevent engagement with CD151?

The authors need to consider an explanation for these discrepancies.

We do not see any discrepancies; our observations are compatible with aspects of both the shedding and the priming model. That PsVs carry HS-cleavage products doesn't imply that HS cleavage is sufficient or required for infection, or that the priming model would be wrong. We do not view our data as being in conflict with the priming model. Most of the above-mentioned papers are now cited.

Altogether, we acknowledge that the study gains importance by directly testing the priming model within our experimental system. We are thankful for the above comments and addressed this issue.

Other issues:

(1) Line 110-111. The statement about PsVs in the ECM being too far away from the cell surface to make physical contact with the cell surface entry receptors is confusing. ECM binding has not been shown to be an obligatory step for in vitro infection.

Not obligatory, but strongly supportive (Bienkowska-Haba et al., Plos Path., 2018; Surviladze et al., J. Gen. Viro., 2015). As recently published by the Sapp lab (Bienkowska-Haba et al., Plos Path., 2018), Direct binding of HPV16 to primary keratinocytes yields very inefficient

infection rates for unknown reasons. Moreover, the paper shows that HaCaT cell ECM binding of PsVs increases the infection of NHEK by 10-fold and of HFK by almost 50-fold.

This idea is referred to again on lines 158-159 and 199. The claim (line 158) that PsV does not interact with the cell within an hour needs to be demonstrated experimentally and seems at odds with multiple laboratories' data. PsV has been shown to directly interact with HSPG on the cell surface in addition to the ECM. Why are these PsVs not detected?

The reviewing editor speculated that HaCaT cells may be a model system in which the in vivo relevant binding to the ECM can be better studied as in non-polarized cell types. This is because binding to the ECM cannot be bypassed by direct cell surface binding. The observation that only few PsVs bind to the basal cell membrane indeed suggests restricted diffusional access of PsVs to binding receptors of the basal membrane. The reviewing editor asked for an experiment showing that more PsVs bind after cell detachment. We performed this experiment and indeed find more PsVs binding to the cell surface of detached cells. This point is very important for the understanding of the study and now we mention it in several sections of the manuscript, as outlined in the following.

Line 125: 'Many PsVs that bind to the ECM may locate distal from the cell surface and are thus unable to establish direct contact with entry receptors. However, they are capable of migrating by an actindependent transport along cell protrusions towards the cell body (Smith et al. 2008; Schelhaas et al. 2008). We aimed for blocking this transport in HaCaT cells, a cell line that is widely used as a cell culture model for HPV infection. HaCaT cells closely resemble primary keratinocytes in key aspects: they are not virally transformed and produce large amounts of ECM that facilitates infection (Bienkowska-Haba et al. 2018; Gilson et al. 2020). In addition, HaCaT cells exhibit cellular polarity that enforces binding of virus particles to the ECM, as the virions cannot bind to receptors/entry components, such as CD151, Itga6 and HSPGs that co-distribute on the basolateral membrane of polarized keratinocytes (Sterk et al. 2000; Cowin et al. 2006; Mertens et al. 1996), making them inaccessible by diffusion.'

Line 205: 'During the CytD incubation, PsVs bind to HSPGs of the basolateral membrane for 5 h. Still, in the cell body area hardly any PsVs are present ($0.14 \text{ PsV}/\mu\text{m}^2$, Supplementary Figure 1B). In the control, the PsV density is several-fold larger (Supplementary Figure 1B). This is expected, as the PsVs bind to the ECM and translocate to the cell body. We wondered whether there are more binding sites at the basal membrane that remain inaccessible to PsVs by diffusion because of the insufficient space between glass-coverslip and basolateral membrane. For clarification, we incubated EDTA detached HaCaT cells in suspension with PsVs for 1 h at 4°C , followed by re-attachment for 1 h. Under these conditions, we find a PsV density 12.4-fold larger than after 5 h of CytD incubation of adhered cells (Supplementary Figure 1B and D). However, it should be noted that these values cannot be directly compared. Aside from the different treatments, another difference lies in the size of the basal membrane, as re-attachment of cells is not complete after only 1 h (compare size of adhered membranes in Supplementary Figure 1A and C). Therefore, the imaged membranes are likely strongly ruffled, which results in the underestimation of the size of the adhered membrane. As a result, we overestimate the PsVs per μm^2 (please note that we cannot re-attach cells for longer times as we would then lose PsVs due to endocytosis). On the other hand, we would underestimate the PsV density at the basal membrane if after re-attachment we image in part also some apical membrane. In any case, the experiment suggests that PsVs bind more efficiently if membrane surface receptors are accessible by diffusion. This is in support of the above notion that the basal membrane may provide more entry receptors than one would expect from the low density of PsVs bound after 5 h CytD (Supplementary Figure 1B). This suggests that under our assay conditions, PsVs cannot easily bypass the translocation from the ECM to the cell body by diffusing directly to the basal membrane. Hence, the large majority of PsVs that enter the cell were previously bound to the ECM. Therefore, HaCaT cells serve as an ideal model for studying the transfer of ECM bound HPV particles to the cell

surface, which is similar to in vivo infection of basal keratinocytes after binding to the basement membrane (Day and Schelhaas 2014; Kines et al. 2009; Schiller et al. 2010; Bienkowska-Haba et al. 2018).’

Line 529: ‘Filopodia usage not only facilitates infection but also increases the likelihood of virions to reach their target cells during wound healing, namely the filopodia-rich basal dividing cells. In fact, several types of viruses exploit filopodia during virus entry (Chang et al. 2016), hinting at the possibility that for HPV and other types of viruses actin-driven virion transport may play a more important role than it is currently assumed. If this is the case, sub-confluent HaCaT cells, or even better single HaCaT cells, would be an ideal model system for the study of these very early infection steps that involve ECM attachment and subsequent filopodia-dependent transport. As shown in Supplementary Figure 1, HaCaT cells have many binding sites for the HPV16 PsVs. However, as they are polarized and the binding receptors are only at the basal membrane, they remain relatively inaccessible by diffusion. Therefore, the ECM binding that is also observed in vivo (Day and Schelhaas 2014) and subsequent transport via filopodia are used upon infection of HaCaT cells that locate at the periphery of cell patches. Here, PsVs bind to the ECM which strongly enhances infection of primary keratinocytes (Bienkowska-Haba et al. 2018). In contrast, HPV can readily bind to HSPGs on the cell surface of nonpolarized cells, and by this bypasses ECM mediated virus priming and the filopodia dependency. We propose that HaCaT cells are a valuable system for studying the very early events in HPV infection that allows for dissecting capsid interaction with ECM resident priming factors and cell surface receptors.’

Finally, please note that in the previous version of the manuscript, we did not question that in many cellular systems PsVs interact with heparan sulfate proteoglycans (HSPGs) present on the cell surface, or both on the cell surface and the ECM. We stated on line 59 While in cell culture virions bind to HS of the cell surface and the ECM, it has been suggested that in vivo they bind predominantly to HS of the extracellular basement membrane (Day and Schelhaas, 2014; Kines et al., 2009; Schiller et al., 2010).’

We hope that after adding the above explanations and the experiment requested by the reviewing editor it is now clear why only few PsVs bind directly (not via the ECM) to the cell surface. We appreciate the reviewer’s and the reviewing editor’s input that has significantly improved the manuscript.

(2) The experiments shown in Figure 5 need to be better controlled. Why is there no HS staining of the cell surface at the early timepoints? This antibody has been shown to recognize N-sulfated glucosamine residues on HS and, therefore, detects HSPG on the ECM and cell surface.

There is staining. However, as the staining at the periphery is stronger and images are shown at the same settings of brightness and contrast, the impression is given that the cell surface is not stained. We have added more images showing HS cell surface staining.

(i) Supplementary Figure 4C shows an enlarged view of the CytD/0 min cell shown in Figure 6A. In the area stained by Itga6, that marks the cell body, HS staining is present, although less abundant in comparison to the ECM.

(ii) In Figure 8, CytD/30 min, a cell is shown with abundant HS in the cell body region (compare cyan and green LUT).

(iii) In newly added Figure 3A, lower panel, another cell with HS in the cell body region is shown.

Please note that the staining is highly variable. We indicate this by stating on Line 373: ‘The pattern of the HS staining (cyan LUT) and the overlap of HS with PsVs and Itga6 are highly variable (Figure 6A).’

Therefore, the conclusion that this confirms HS coating of PsV during release from the ECM (line 430431) is unfounded. How do the authors distinguish between "HS-coated virions" and HSPG-associated virions?

The transient increase in the PCC at CytD/30 min can be interpreted as PsV/HS co-transport or as direct binding of PsVs to cell surface HSPGs. However, two arguments support co-transport.

First, we find that CytD/PsVs increases the HS intensity (see newly added Figure 3, confirming old Figure 5 that is now Figure 6). We state on line 290 ‘... that without actin-dependent PsV translocation HS cleavage products are retained in the ECM, consistent with the hypothesis that cleaved HS remains associated with PsVs (Ozbun and Campos 2021).

Second, the distance between HS and Itga6 (the cell body marker) decreases over time after CytD removal, which suggests movement of HS to the cell body (Supplementary Figure 8D). We state on line 422: ‘The movement of HS towards the cell body after removal of CytD, which indirectly demonstrates that PsVs are coated with HS, is suggested by a shortening of the HS-Itga6 distance over time (Supplementary Figure 8D).’

It is difficult to comprehend how the addition of 50 vge/cell of PsV could cause such a global change in HS levels.

Some areas are covered with confluent cells, to which hardly any PsVs are bound, because accessing their basolateral membrane is nearly impossible, and PsVs do not bind to the exposed apical membrane as well. We assume this is a major difference to cultures of unpolarized cells, where PsVs should distribute more or less equally over cells. This means that in our experiments the vge/cell is not a suitable parameter for relating the magnitude of an effect to a defined number of PsVs. In the ECM, the PsV density is very high, enabling one cell to collect, in theory, several hundred PsVs, much more than expected from the 50 vge/cell.

We state on line 135: ‘Frequently, we observe patches of confluent cells which are common to HaCaT cells. Cells at the center of these patches are dismissed during imaging, because there are no anterogradely migrating PsVs at these cells. A second reason for our dismissal of these cells is that hardly any PsVs are bound to them, possibly because their basal membranes are inaccessible by diffusion. Instead, we focus on isolated HaCaT cells or cells at the periphery of cell patches. In these cells, we find more PsVs per cell than one would expect from the employed 50 viral genome equivalents (vge) per cell, indicating that PsVs are unequally distributed between the cells.’

The claim that the HS levels are decreased in the non-cytochalasin-treated cells due to PsV-induced shedding needs to be demonstrated.

We did not claim that PsVs induce shedding, we rather believe they retain shedded HS. Without PsVs, the shedded HS is washed off from the ECM. We have reproduced the observation made in old Figure 5 (now Figure 6) in the newly added Figure 3 that also shows that PsVs alone have no effect on the HS intensity, only when present together with CytD. We state on line 277: ‘As outlined above, during the 5 h incubation with CytD, proteases in the ECM are expected to cleave HS chains. These cleavage products should be able to diffuse out of the ECM, unless they remain associated with nontranslocating PsVs. In the control, PsV associated HS cleavage products would leave the ECM through PsV translocation.... Using an antibody that reacts with an epitope in native heparan sulfate chains, only after CytD and if PsVs are present, the level of HS staining is significantly increased (Figure 3B). As shown in Figure 3A, stronger HS staining at PsVs (open arrows) and as well in PsV free areas (closed arrows) was observed... Collectively, our findings indicate that without actin-dependent PsV translocation HS cleavage products are retained in the ECM, consistent with the hypothesis that cleaved HS remains associated with PsVs (Ozbun and Campos 2021).’

If HS is actually shed, staining of the cell periphery could increase with the antibody 3G10, which detects the HS neoepitope created following heparinase cleavage.

We have tested the antibody by which we obtain only a very weak staining (Supplementary Figure 2), not allowing to differentiate between an increase in the cell periphery and the cell body area. We still include the experiment as it suggests that CytD has no effect on HS processing. We state on line 286: ‘As additional control and shown in Supplementary Figure 2, we use an antibody that reacts with a HS neo-epitope generated by heparitinase-treated heparan sulfate chains (Yokoyama et al. 1999; for details see methods). This neo-epitope staining is independent of the presence of CytD and the incubation time, suggesting that CytD does not directly affect HS processing.’

Reviewer #2 (Public review):

Summary:

Massenberg and colleagues aimed to understand how Human papillomavirus particles that bind to the extracellular matrix (ECM) transfer to the cell body for later uptake, entry, and infection. The binding to ECM is key for getting close to the virus's host cell (basal keratinocytes) after a wounding scenario for later infection in a mouse vaginal challenge model, indicating that this is an important question in the field.

Strengths:

The authors take on a conceptually interesting and potentially very important question to understand how initial infection occurs in vivo. The authors confirm previous work that actin-based processes contribute to virus transport to the cell body. The superresolution microscopy methods and data collection are state-of-the art and provide an interesting new way of analysing the interaction with host cell proteins on the cell surface in certain infection scenarios. The proposed hypothesis is interesting and, if substantiated, could significantly advance the field.

Weaknesses:

As a study design, the authors use infection of HaCaT keratinocytes, and follow virus localisation with and without inhibition of actin polymerisation by cytochalasin D (cytoD) to analyse transfer of virions from the ECM to the cell by filopodial structures using important cellular proteins for cell entry as markers.

First, the data is mostly descriptive besides the use of cytoD, and does not test the main claim of their model, in which virions that are still bound to heparan sulfate proteoglycans are transferred by binding to tetraspanins along filopodia to the cell body.

The study identifies a rapid translocation step from the ECM to CD151 assemblies. We have no data that demonstrates a physical interaction between PsVs and CD151. In the model figure, we draw CD151 as part of the secondary receptor complex. We are sorry for having raised the impression that PsVs would bind directly to CD151 and have modified the model Figure accordingly. In the new model figure (Figure 9), the first contact established is to a CD151 free receptor.

Second, using cytoD is a rather broad treatment that not only affects actin retrograde flow, but also virus endocytosis and further vesicular transport in cells, including exocytosis. Inhibition of myosin II, e.g., by blebbistatin, would have been a better choice as it, for instance, does not interfere with endocytosis of the virus.

As we focus on early events, we are not concerned about CytD blocking as well late steps in the infection cascade, like endocytosis. However, we agree that a comparison between CytD

and blebbistatin would be very interesting. We added Figure 8, showing that blebbistatin only partially stops migration.

Line 429: ‘Actin retrograde transport, which underlies the here observed virion transport, is the integrative result of three components (Smith et al. 2008; Schelhaas et al. 2008).... As CytD broadly interferes with F-actin dependent processes, we investigated the effects upon inhibition of only one of the three components, namely the myosin II mediated retrograde movement towards the cell body. Instead of CytD, we employed in the 5 h preincubation the myosin II inhibitor blebbistatin. For the control (0 min), we show in Figure 8A one example of a cell with comparatively many PsVs at the periphery (as mentioned above, the PsV pattern is highly variable) to better illustrate the difference to the PsV pattern occasionally seen with blebbistatin. After blebbistatin treatment (0 min), PsVs are still distal to the cell body but less dispersed than after CytD treatment, seemingly as if translocation started but stopped in the midst of the pathway (Figure 8A, blebbistatin). The PCC between PsVs and HS, like after CytD (Figure 6C), is elevated after blebbistatin, albeit the effect is not significant (Figure 8C). The cell body PCC, is not at 30 min (CytD) but already at 0 min elevated (compare Figure 6D to Figure 8D), which can be explained by partial translocation. This is further supported by the fact that only 8% of PsVs are closely associated with HS (Figure 8E; blebbistatin, 0 min) compared to 15% after CytD treatment (Figure 6E; 0 min). Furthermore, after 0 min PsV incubation with blebbistatin we observe no effect on the HS intensity (compare Figure 8B to Figure 3B and Figure 6B). Hence, in contrast to CytD, blebbistatin does not trap the PsVs in the ECM where they associate with HS, but ongoing actin polymerization pushes actin filaments along with PsVs towards the cell body.’

Third, the authors aim to study transfer from ECM to the cell body and the effects thereof. However, there are substantial, if not the majority of, viruses that bind to the cell body compared to ECM-bound viruses in close vicinity to the cells.

Please see our detailed reply to referee #1 that has raised the same issue. In brief, we agree that in multiple cell culture systems viruses bind preferentially to the cell surface directly. However, in HaCaT cells, the majority of PsVs does not bind directly to the basal membrane but gets there after initial binding to the ECM. Thus, we believe our system appropriately models the physiologically relevant scenario of ECM-to-cell transfer, as also speculated by the reviewing editor that has suggested an experiment showing that more PsVs bind to detached cells (please see above).

This is in part obscured by the small subcellular regions of interest that are imaged by STED microscopy, or by the use of plasma membrane sheets. As a consequence, the obtained data from time point experiments is skewed, and remains for the most part unconvincing due to the fact that the origin of virions in time and space cannot be taken into account. This is particularly important when interpreting association with HS, the tetraspanin CD151, and integral alpha 6, as the low degree of association could originate from cell-bound and ECM-transferred virions alike.

As already stated above, we observe massive binding of PsVs to the ECM, in contrast to very few PsVs that diffuse beneath the basolateral membrane of the polarized HaCaT cells and do bind directly to the cell surface. In other cellular systems, cells may hardly secrete ECM, are not polarized, and therefore virions can easily bypass ECM binding. Therefore, it is reasonable to assume that in HaCaT cells the large majority of PsVs found on the cell body originates from the ECM.

Fourth, the use of fixed images in a time course series also does not allow for understanding the issue of a potential contribution of cell membrane retraction upon cytoD treatment due to destabilisation of cortical actin. Or, of cell spreading upon cytoD washout.

The newly added blebbistatin experiment suggests that the initial translocation is exclusively dependent on retrograde actin flow. However, we agree that we are not able to unravel more details regarding the different possible contributions to the movement. Importantly, the lack of PCC increase after CytD/leupeptin removal (Figure 2D) suggest there is not much cell spreading into the area of accumulated PsVs. Please see our more detailed reply to the same issue raised by the same referee in the recommendations for the authors.

The microscopic analysis uses an extension of a plasma membrane stain as a marker for ECM-bound virions, which may introduce a bias and skew the analysis.

The dye TMA-DPH stains exclusively cellular membranes and not the ECM. The stain is actually used to delineate the cell body from the ECM area (please see Figure 1).

Fifth, while the use of randomisation during image analysis is highly recommended to establish significance (flipping), it should be done using only ROIs that have a similar density of objects for which correlations are being established.

We agree that the way of how randomization is done is very important. Regarding the association of PsVs with CD151 and HS, we corrected for random background association, which is now explained in more detail in the Figure legend of Supplementary Figure 7: “On flipped images, we often find values more than half of the values of the original images, demonstrating that many PsVs have a distance ≤ 80 nm to CD151 merely by chance (background association)... (C) Each time point in (A) and (B) obtained from flipped images is the average of three biological replicates. We use these altogether 24 data points, plotting the fraction of closely associated PsVs against the CD151 maxima density. The fraction increases with the maxima density, as the chance of random association increases with the maxima density. The fitted linear regression line describes the dependence of the background association from the maxima density. As a result, the background association (y) can be calculated for any maxima density (x) in original images with the equation $y = 2.04x$. Please note that the CytD/0 min may be overcorrected as we subtract background association with reference to the CD151 maxima density of the entire ROI (for an example ROI see Supplementary Figure 6A), although the local maxima density at distal PsVs is lower. On the other hand, PsVs at the cell border may have a larger local CD151 maxima density and consequently are undercorrected.’

For instance, if one flips an image with half of the image showing the cell body, and half of the image ECM, it is clear that association with cell membrane structures will only be significant in the original.

We are aware of this problem. For instance, it would produce ‘artificially’ low PCCs after flipping images of PsV/HS stainings (please see negative PCC value after flipping in Supplementary Figure 8). In this case, we do not use as argument that in flipped images the PCC is lower. Instead, we would argue that over time the PCC changes in the original images. We still provide the PCC values of flipped images, as additional information, showing that in most cases we obtain after flipping a PCC of zero, as expected

Hence, we fully agree that careful controls in image analysis is required, and used the above-described method for the correction of background association when the fraction of closely associated PsVs is analyzed. We do not use a lower PCC value in flipped images as argument if not appropriate.

I am rather convinced that using randomisation only on the plasma membrane ROIs will not establish any clear significance of the correlating signals.

Figure 6D and 8D show the PCC specifically of the cell body (only of plasma membrane ROIs). In flipped images (not shown in the previous version for clarity), we obtain significantly

lower PCCs (Supplementary Figure 8F/G and Supplementary Figure 10C/D). We propose that in this case it would be appropriate to use a lower PCC of flipped images as argument for specific association. Still, also in this experiment we argue with a change in the PCC over time, and not with a PCC of zero after flipping. As above, we still provide the PCC values of flipped images as additional information.

| *Also, there should be a higher n for the measurements.*

One replicate is based on the average of 14-15 cells for each condition (more for figure 4). Hence, in a typical experiment (Control and CytD with 4 time points) about 120 cells are analyzed, which is a broad basis for the averages of one replicate.

We realize that with three biological replicates we find significant effects only if we have strong effects or moderate effects with very low variance.

Recommendations for the authors:

Reviewing Editor:

The focus on the events of HPV infection between ECM binding and keratinocyte-specific receptor binding is unique and interesting. However, I agree with the reviewers that some of the conclusions could use more experimental support, as detailed in their comments. The failure to detect direct binding of the PsV to HSPGs on the cell surface in in vitro assays contradicts much of the published literature. For example, others have found that HPV capsids bind cultured cell lines in suspension, i.e., in the absence of ECM. Do EDTA-suspended HaCaT cells bind PsV? Is the binding HSPG dependent? If the authors think that failure to detect direct cell binding of HaCaTs is an unusual feature of these cell lines or culture condition,s then it would be helpful to provide an explanation. However, it is worth noting that an in vitro system where the cells do not directly bind capsids through HSPG interactions would be a much better model for studying the stages of HPV infection that are the focus of this study, since there is no direct binding of keratinocytes in vivo.

We are thankful for this comment that had a strong influence on the revision. The suggested experiment has been incorporated as new Supplementary Figure 1. It shows that many more PsVs bind to the cell surface of cells in suspension than to adhered cells. As suggested by the reviewing editor, we explain now that HaCaT cells are a suitable model system for studying the in vivo transport from the ECM to the cell body that in these cells, due to their polarization, cannot be bypassed (for more details please see our replies above addressing these issues).

| *Because conclusions drawn regarding HS interactions are largely based on experiments using a single HS mAb, it is important that the specificity of this mAb is described in more detail, either based on the literature or further experimentation.*

We provide now detailed information about the HS antibodies used in the study. We state on line 282 ‘Using an antibody that reacts with an epitope in native heparan sulfate chains...’ and on line 286 ‘we use an antibody that reacts with a HS neo-epitope generated by heparitinase-treated heparan sulfate chains...’ and in the methods section ‘For Heparan sulfate (HS) a mouse IgM monoclonal antibody (1:200) (amsbio, cat# 370255-S) was used that reacts with an epitope in native heparan sulfate chains and not with hyaluronate, chondroitin or DNA, and poorly with heparin (mAb 10E4 (David et al., 1992)). For HS neo-epitope (Yokoyama et al., 1999) detection, a mouse monoclonal antibody (1:200) (amsbio, cat#370260-S) was used that reacts only with heparitinase-treated heparan sulfate chains, proteoglycans, or tissue sections, and not with heparinase treated HSPGs. The antibody recognizes desaturated uronic acid residues (mAb 3G10 (David et al., 1992)).’

Reviewer #1 (Recommendations for the authors):

(1) The phrase "tight association" or similar is repeatedly used and is not acceptable for microscopic studies; use "close association", which has no affinity connotations.

Has been changed as suggested by the referee.

(2) Why are lysine-coated coverslips used for microscopy? HaCaT cells adhere tightly to untreated glass, and this coating could affect the distribution of ECM and extracellular PsV.

We believe a tight association of the basal cell membrane to its substrate, as in vivo, where the basal membrane is tightly adhered to other cells, is important in these experiments. In weakly adherent cells more PsVs may bind to the cell surface, bypassing the transport step. Hence, although HaCaT cells may not require the coat and would be able to adhere to glass, the association may not be tight enough to mimic in vivo conditions.

(3) What is the reason to use detection of the pseudogenome for some of the experiments instead of L1 detection throughout? The process of EdU detection is sufficiently denaturing to affect some protein epitopes. The introduction of this potential artifact doesn't seem warranted for capsid detection experiments.

The L1 and the Itga6 antibody are from the same species, wherefore we have used in Figures 4 and 6 click-labeling of the reporter plasmid. We do not disagree with the notion of the referee, that EdU detection may denature the epitope of some proteins. For instance, we have observed a different staining pattern for CD151; for Itga6 and HS we saw no obvious difference in the staining patterns. In double staining experiments using L1 antibody and click-labeling, both staining patterns overlapped very well, indicating that click-labeling is suitable to visualize PsVs.

(4) What concentration of TMA-DPH was used?

TMA-DPH is a poorly water-soluble dye that becomes strongly fluorescent upon insertion into a membrane. Because of its poor water solubility, a precise concentration cannot be given. We added 50 µl of a saturated TMA-DPH solution in PBS to 1 ml of PBS in the imaging chamber. We state this now in the methods section.

(5) Line 419: This statement is misleading. Although PsV interaction with HSPG on the ECM is crucial for infectious transfer to cells, the majority of the PsV binding on the ECM has been attributed to interaction with laminin 332. Treatment of PsV with heparin causes sequestration to the ECM.

We are sorry for the confusion and have removed the misleading statement.

(6) Some reference choices are poor:

Line 54: Ozbun and Campos, this is not the correct reference

In the review we cited, in the introduction it is stated that PsVs establish infection via a break in the epithelial barrier? However, we have replaced this reference by a review that focuses more on epithelial wounding: 'Ozbun, Michelle A. (2019): Extracellular events impacting human papillomavirus infections: Epithelial wounding to cell signaling involved in virus entry. In Papillomavirus research (Amsterdam, Netherlands) 7, pp. 188–192. DOI: 10.1016/j.pvr.2019.04.009.'

Line 2012: Doorbar et al., this is not the correct reference.

Thank you for pointing this out (..we assume the referee refers to line 104 and not line 2012). We have noticed this error during revision. As it is difficult to get a specialized review on this topic, we now cite Ozburn and Campus, 2021 that states PsVs are ‘structurally and immunologically indistinguishable from lesion- and tissue-derived HPVs.’

Minor issues:

(1) It is difficult to appreciate the ECM and cell surface binding pattern from the provided images, which do not even contain an entire cell. We need to see a few representative field views with the ECM delineated with laminin 332 staining, as HS antibodies stain both the ECM and cell surface.

We now provide overview images in Supplementary Figure 4. The only experiment requiring a clear delineation between ECM and cell surface is the experiment of Figure 4. Here, we do not use the HS as a reference staining because it stains both the ECM and the cell surface.

(2) For Figure 1E, the cells were only infected for 24 hours. The half-time for infectious internalization of HaCaT cells was shown to be 8 hours for cell-associated PsV and closer to 20 hours for PsV that was associated with the ECM prior to cell association (Becker et al., 2018). Why was such a short infection time chosen?

During assay establishment it has been observed that after 24 h the luciferase activity is optimal.

(3) Figure 5, the staining of uninfected cells +/- cyto treatment needs to be included.

Now visible in new Figure 3.

I am confused by lines 54-57. It seems as if the authors are claiming that HSPGs are not present on the ECM. This sentence, as written, is misleading.

We agree, and state now on line 58 ‘Here, virions bind to the linear polysaccharide heparan sulfate (HS) that is present in the extracellular matrix (ECM) but as well on the plasma membrane surface. HS is attached to proteins forming so called heparan sulfate proteoglycans (HSPGs).’

Reviewer #2 (Recommendations for the authors):

There are further issues that are not pertaining to the study design that I find important.

(1) It remains speculative whether the virions that are transferred from the ECM are actually structurally modified.

The newly added Figure 2, showing that leupeptin blocks infection in our assay, suggests that virions indeed are primed.

(2) The origin of HS correlated with virions on the cell body after transfer is also not clear: does the virus associate with cell surface HS, or does it bring HS from the ECM? Simply staining HS against N-sulfated moieties does not allow such conclusions.

This issue has been already raised in the public review to which we replied above. In brief, we agree that the transient increase of the PCC between PsVs and HS in the cell body region can be also explained by PsVs coming from the ECM without HS and binding to cell surface HS, or from PsVs binding directly (not via the ECM) to cell surface HSPGs. However, there are two more arguments indicating that PsVs are coated with HS. Please see our detailed reply above.

(3) Figure 1: There are few, if any, filopodia in untreated cells. It would be good to quantify their abundance to substantiate that resting HaCat cells are indeed a good model for filopodial transport bs. membrane retraction / spreading. In HaCat ECM, the virus also binds to laminin-332 for a good part. Would this not also confound the analysis?

At first glance, the number of filopodia appears to be too low to account for such an efficient transport. However, please note that the formation of filopodia is very dynamic, and that they can form and disappear within minutes (see below). We also often observe many PsVs aligned at one filopodium. Moreover, not every cell periphery exhibits large accumulations of PsVs. Therefore, we believe it is in principle possible that filopodia are largely responsible for the transport. We cannot exclude that we overestimate the transport rate due to partial cell spreading after CytD removal, which, however, we consider as rather unlikely as in Figure 2 we observe no increase in the PCC when leupeptin was present during the CytD incubation. Under these conditions, PsVs do not translocate but cells could spread, and this would increase the PCC between PsVs and F-actin if cells would spread into the area of accumulated PsVs.

We now state on line 304: ‘This suggests that the half-time of PsV translocation from the periphery to the cell body is about 15 min. In fact, the half-time maybe longer, as we cannot exclude that cell spreading after CytD removal contributes to less PsVs measured in the cell periphery,’ and on line 477 ‘As mentioned above, the half-time could be longer if cell spreading is in part responsible for the translocation of PsVs onto the cell body. However, we assume that this is rather unlikely, as cell spreading would increase the PCC between PsVs and F-actin under a condition where filopodia mediated transport is blocked but not cell spreading, which is not the case (Figure 2B and D, CytD/leupeptin).’

(4) Figure 2: This would benefit from live cell analysis. There are considerable amounts of virions on the cell body, which partially contradicts statements from Figure 1.

Does the referee refer to the images shown in Figure 4 (old Figure 2)? Please note that at CytD/0 min there are hardly any PsVs in the cell body region, the fluorescence (magenta LUT) is autofluorescence (this is explained in the results section). Only at later time points PsVs are in the cell body region.

The fast transfer to the cell body after cyto D washout is based on the assumption that filopodia formation and transport along them (and not membrane extension) occur quickly. Is this reasonable?

We are no experts on filopodia, but one finds references suggesting that they grow at rates of several μm per minutes and have lifetimes between a few seconds and several minutes. Hence, within the 15 min we determine for the transport, cells may need a few minutes to recover from CytD, a few minutes to form filopodia that reach out into the ECM, and a few minutes for the transport itself. However, we agree that we cannot exclude membrane extension contributing to our observed transport, although we consider this as rather unlikely (see above).

(5) Figure 3: The rationale of claiming the existence of 'endocytic structures' needs to be better explained and quantified in the according supplementary figure.

We now state in the legend ‘We propose that the agglomerated CD151 maxima close to PsVs feature the characteristics of endocytic structures, as CD151 has been shown to co-internalize with PsVs (Scheffer et al. 2013), and as these structures invaginate into the cell, like PsV filled tubular organelles previously described by electron microscopy (Schelhaas et al. 2012).’ For a proper quantification of these highly variable structures a much larger sample would be required.

The formation of virus-filled tubules upon cytoD treatment has been previously reported. Are these viruses that come from the cell body or from the ECM?

With the new data and explanations that have been added to the manuscript, it should be clear that it is reasonable to assume that they come largely from the ECM.

(6) Figure 4: How are the subcellular ROIs chosen? Is there not a bias by not studying a full cell?

We now explain better how we chose cells for analysis. We state on line 138 ‘Instead, we focus on isolated HaCaT cells or cells at the periphery of cell patches. In these cells, we find more PsVs per cell than one would expect from the employed 50 viral genome equivalents (vge) per cell, as PsVs are unequally distributed between the cells. Moreover, these PsVs usually are not homogeneously distributed around the cell but concentrate at one region. We investigate the translocation of PsVs from these regions, defining ROIs for analysis that cover PsVs at the periphery and the cell body (see Supplementary Figures 6A and 8A).’

(7) Figure 5/6: The data needs a better analysis on correlation by using randomisation as explained above.

Please see our reply to the same point of the public review raised by the same referee.

(8) Figure 7: This model involves CD151 being a mediator in transfer, but this has not been functionally shown. There are HaCaT CD151 KO cells available (from the Sonnenberg lab), it would be good to use those to test the model and whether transfer indeed involves CD151.

As already stated above, we are sorry for having raised the impression that PsVs bind directly to CD151. The model Figure has been modified. Please see our reply above.

(9) The manuscript would benefit from a number of experiments addressing the most crucial issues:

(a) As mentioned before, the use of blebbistatin, which blocks myosin II function and arrests actin retrograde flow within seconds of addition, would be a good inhibitor to control for transfer in at least some of the most crucial experiments.

In Figure 8 we have tested blebbistatin. Please see our reply above.

(b) Live cell analysis would allow for monitoring of whether membrane retraction upon cytoD treatment would have to be taken into account for the analysis of the data. The same is true for the cytoD washouts, upon which most cells exhibit pronounced membrane spreading. The latter is important to support filopodial transport rather than membrane ruffling and spreading, leading to the clearance of extracellular virions from the ECM.

We agree that this would be desirable. As replied above, we now discuss the issue of possible membrane spreading and reason why we consider it as rather unlikely.

(c) To rid oneself of the issue of plasma membrane-bound virions as a confounding factor, one could use cells treated by sodium chlorate, which leads to undersulfation of HS on the cell surface, and seed them onto ECM with functional HSPGs. This would then indeed establish that the HS and virus are transferred together.

We agree that this would be a smart experiment. As the main focus of our study is not clarifying whether PsVs are coated with HS or not, we gave other experiments priority.

(10) The manuscript is, while carefully and thoughtfully worded on the issue of microscopy analysis, for a good part, extrapolating too strongly from the authors' data and unsubstantiated assumptions to conclude on their model. It would be good if the authors would support their claims with previous or their own experimental work. Just two examples of several: the assumption that cell-bound virions are negligible should be substantiated, as the literature would indicate otherwise.

We determined the PsV density in adhered, CytD treated cells, and find around 0.14 per μm^2 (Supplementary figure 1B), which is 4 to 5-fold less when compared to the PsV density quantified in an area covering the cell body and the periphery (Figure 1B, see line 174 for PsVs/ μm^2 values). Quantifying the PsV density only in the periphery would yield a severalfold larger difference. However, due to the limited resolution of the microscope we would strongly underestimate the PsV density in the accumulations. We prefer not to discuss this in detail, as exact numbers are difficult to obtain.

Line 129: Cyto D should not inhibit the enzymes modifying HS or proteins (including virions). This is true, but cytoD may limit their secretion and abundance.

We show in Figure 3 that CytD does not reduce HS staining (e.g., by limiting HS secretion, as suggested by the referee), suggesting that it rather does not limit secretion.

We thank the referees and the reviewing editor for their helpful comments!

<https://doi.org/10.7554/eLife.107139.2.sa0>